Quantifying Market Greeks in Soccer Betting:

A Theoretical Framework for In-Play Price Sensitivity

Adhitya Suppiah

Syracuse University

Spring 2026

Abstract

This thesis adapts the framework of financial derivatives — specifically Delta, Theta, and Gamma — to in-play soccer betting on the Betfair Exchange. Using a proprietary dataset of 791 matches and 1,465 high-confidence goal events across five major European competitions, I estimate the three core price sensitivities empirically. Underdog goals produce price movements approximately twice as large as favorite goals; draw probability accelerates 4.5× from early to late play in scoreless matches; and Gamma peaks in close late-match situations, where median Delta reaches 0.531. Markets are aggregately efficient but exhibit a counterintuitive overshoot pattern that is larger for favorite than underdog goals.

Introduction

On any given match day across Europe's top soccer leagues, millions of dollars change hands on betting exchanges in real time as goals are scored, cards are shown, and the clock ticks toward full time. Behind every price movement on these exchanges lies a mechanism strikingly similar to what drives options markets: an asset (a betting position) whose value depends on a stochastic underlying process (the match score) that unfolds over a finite time horizon. And yet, despite the structural parallel between in-play betting and financial derivatives, no prior empirical work has systematically measured the price sensitivities of in-play betting markets using the tools of derivatives theory.

This thesis bridges that gap. By adapting the framework of financial derivatives theory—specifically the concepts of Delta (Δ), Theta (Θ), and Gamma (Γ)—to in-play soccer betting on the Betfair Exchange, this research provides the first systematic measurement of how live betting prices respond to goals, to time, and to changing match states. Using a proprietary dataset of 791 matches spanning England's Premier League, Spain's La Liga,

Italy's Serie A, Germany's Bundesliga, and the UEFA Champions League—with 1,465 high-confidence in-play goal events captured via the Betfair Stream API at 68 percent recall—I estimate the three core price sensitivities empirically and test a series of hypotheses about their structure.

The motivation for this research is both theoretical and practical. From a theoretical perspective, soccer betting markets present a unique laboratory for studying price formation in time-constrained environments with discrete information shocks. Unlike equity options, where the underlying asset moves continuously and the time horizon is defined by contract expiry, soccer betting features discrete state changes (goals, red cards) against a hard 90-minute clock. This creates a natural analogue to barrier options and jump-diffusion processes: Delta measures the price jump at each goal, Theta captures the continuous decay of draw probability between goals, and Gamma measures the market's sensitivity to that very jump—the second-order effect that determines how quickly price sensitivity itself is changing.

From a practical standpoint, understanding the Greeks of betting markets has direct applications for market participants. Professional bettors and market makers on exchanges like Betfair need to understand how prices will respond to match events in order to manage risk effectively. A quantified Gamma surface, for example, tells a trader exactly when late-game close matches become most explosive—information that is essential for sizing positions and setting limits. Similarly, Theta estimates allow traders holding draw positions to understand precisely how rapidly the time value of that position erodes over the course of a scoreless match.

This analysis is organized as follows. First, I describe the data sources and the process of constructing the analytical dataset, including the synchronization of market price data with match event timestamps. Second, I present summary statistics characterizing the

sample of matches, market microstructure, and baseline Greek estimates. Third, I present pre-model visualizations that illustrate the key patterns in the data. Fourth, I present three formal empirical models—one for each Greek—and report their results. Fifth, I present market reaction speed and behavioral anomaly analyses. Sixth, I discuss the implications of these findings for market participants, behavioral economics, and the broader literature on prediction market efficiency. I conclude with directions for future research.

Literature Review

This thesis sits at the intersection of financial derivatives theory, market efficiency, behavioral finance, and sports analytics. The literature reviewed below establishes the theoretical foundations on which the empirical Greeks framework rests and identifies the specific gaps that this thesis fills. The review proceeds from the most general theoretical foundations—options pricing and jump-diffusion processes—through the relevant probability models for soccer scoring, the empirical literature on betting market efficiency and behavioral anomalies, and finally the microstructure considerations that govern price formation on the Betfair Exchange.

Options Pricing Theory and the Greeks

The theoretical foundation for this thesis derives from Black and Scholes' (1973) seminal model for pricing European options. Their key insight—that a derivative's price could be uniquely determined through a dynamic hedging strategy involving the option and the underlying asset—produced the Black–Scholes partial differential equation and a closed-form pricing formula. Merton (1973) extended this framework, providing rigorous mathematical foundations using stochastic calculus and earning the 1997 Nobel Prize alongside Scholes (Black having passed away in 1995).

The model's enduring legacy lies less in its pricing formula than in the analytical framework it established for measuring price sensitivities—the “Greeks.” Delta (Δ) measures the sensitivity of option price to a one-unit change in the underlying; gamma (Γ) captures the rate of change of delta itself, reflecting the convexity of the option's price function; theta (Θ) quantifies the erosion of option value as time passes; and vega measures sensitivity to changes in implied volatility. Hull (2018) provides the standard reference, demonstrating that the Greeks are not mathematical curiosities but essential tools for risk management, hedging, and identifying relative value opportunities.

Critical for this thesis is the time-dependent behavior of the Greeks. Theta exhibits non-linear acceleration as expiration approaches: for at-the-money options, theta increases dramatically in the final days before expiry, while gamma similarly spikes (Natenberg, 2015). Cox, Ross, and Rubinstein (1979) introduced the binomial options pricing model, which discretizes time into a finite number of steps; this discrete-time approach offers a useful conceptual bridge to soccer betting, where the match unfolds through 90 discrete minutes and the “underlying” (the scoreline) can only change by integer increments.

Thesis connection. The Greeks framework provides this thesis with its core analytical apparatus. Just as Delta, Theta, and Gamma quantify equity-option sensitivities to underlying price, time, and convexity, this thesis empirically estimates analogous sensitivities of Betfair Exchange prices to goals (Δ), elapsed match minute (Θ), and the joint state of time and score (Γ).

Jump-Diffusion Models and Discrete Event Pricing

While the Black–Scholes framework assumes continuous geometric Brownian motion, real-world asset prices often exhibit sudden, discontinuous movements. Merton (1976) addressed this limitation by superimposing a compound Poisson jump process onto the

standard Brownian motion. His model introduced three additional parameters—jump intensity, mean jump size, and jump-size standard deviation—and produced a semi-closed-form solution as an infinite series of Black–Scholes prices weighted by Poisson jump probabilities.

Kou (2002) extended this framework with a double exponential jump-diffusion model that maintained analytical tractability while producing leptokurtic return distributions and the "volatility smile" observed empirically. The double exponential specification produces asymmetric jump distributions—different magnitudes for upward and downward jumps—aligning well with soccer betting, where goals by favorites and underdogs produce structurally different price impacts despite each representing a single discrete event. Bates (1996) demonstrated that jump risk is often systematic rather than diversifiable and that combining stochastic volatility with jump processes produces superior empirical fits to options data than either feature alone.

Thesis connection. In-play soccer betting markets exhibit precisely the dual structure—continuous drift punctuated by discrete shocks—that jump-diffusion models capture. Goals, red cards, and penalties function as Poisson-distributed jumps producing immediate discontinuous probability movements; between events, prices drift continuously as time passes and minor information accrues. This thesis treats Betfair price evolution as an empirical realization of a jump-diffusion process and measures its Delta (jump magnitude), Theta (continuous drift), and Gamma (sensitivity acceleration) directly.

Poisson Models for Soccer Goal Scoring

Maher (1982) established the foundational framework for modeling soccer scores as Poisson-distributed random variables, parameterizing each team's scoring rate as a

function of attacking strength, opponent defensive weakness, and home advantage. He found that an independent Poisson model adequately described English Football League scores, with a small positive correlation (~0.2) between home and away goals.

Dixon and Coles (1997) substantially refined this approach in a landmark paper. They identified two limitations of the basic independent Poisson model: systematic underestimation of low-scoring outcomes (0–0, 1–0, 0–1, 1–1) and static treatment of team strengths over time. They introduced a correction parameter ρ that modified the joint probability of low-scoring outcomes and an exponentially weighted likelihood function—$\varphi(t) = \exp(-\xi t)$—that down-weighted older results to capture form changes. Crucially, they demonstrated that their model could identify profitable betting opportunities, establishing a direct link between Poisson-based scoring models and market inefficiency.

Dixon and Robinson (1998) extended the framework with a birth-process model that explicitly incorporated in-play dynamics. Goals were treated as interacting birth processes whose scoring rates evolved throughout the match and depended on the current scoreline, with overall rates tending to rise over the 90 minutes. Subsequent work in this tradition has documented score-state-dependent scoring patterns, often interpreted as the product of tactical risk-taking by trailing teams and defensive consolidation by leading teams. Karlis and Ntzoufras (2003) proposed a bivariate Poisson model capturing positive score correlation through a latent common Poisson variable. Koopman and Lit (2015) introduced a dynamic bivariate Poisson framework using state-space methods, and Baio and Blangiardo (2010) adopted a Bayesian hierarchical approach with MCMC estimation, allowing principled uncertainty quantification.

Thesis connection. The Poisson framework provides this thesis with its theoretical benchmark. The empirical Theta in scoreless intervals is compared against the theoretical

rate calibrated at $\lambda = 1.16$ goals per team per game; deviations from this benchmark identify the role of soft information beyond what simple Poisson dynamics capture. The Dixon–Robinson finding of state-dependent scoring rates also informs the score-state heterogeneity observed empirically in the Gamma surface.

The Efficient Market Hypothesis and Prediction Markets

Fama (1970) formalized the efficient market hypothesis (EMH), defining an informationally efficient market as one in which prices fully incorporate all available information, and distinguished three forms of efficiency: weak, semi-strong, and strong. The semi-strong form is most relevant to in-play betting, where prices should update immediately and fully in response to observable match events.

Levitt (2004) argued that betting markets offer several advantages over traditional financial markets for testing the EMH: unambiguous terminal values, clearly defined expiration dates, and relatively transparent information environments. Fama (1998) cautioned that apparent anomalies are often fragile, sensitive to methodology, and frequently disappear in out-of-sample tests—a caution equally applicable to claims of inefficiency in betting markets where transaction costs can erode apparent mispricings. Wolfers and Zitzewitz (2004) reviewed prediction markets broadly, concluding that prices are generally well-calibrated probability forecasts but not immune to behavioral biases such as the favorite-longshot bias. The conceptual foundation traces to Hayek (1945), who argued that the price system's fundamental function is to aggregate dispersed knowledge that no single agent possesses in its entirety.

Thesis connection. This thesis treats Betfair Exchange prices as an aggregated probability forecast that should respond rapidly and unbiasedly to information shocks. The 90% within-300-second time-to-equilibrium and the absence of a kickoff favorite-

longshot bias confirm semi-strong efficiency at the aggregate level, while systematic post-goal overshoot identifies behavioral micro-dynamics that the standard EMH cannot accommodate.

In-Play Betting Market Efficiency and Price Discovery

The Betfair Exchange operates as a continuous double auction—structurally similar to major financial exchanges—where participants submit limit and market orders at self-selected prices. The exchange charges a commission on net winnings (typically 2–5%) rather than embedding a margin in the odds, producing tighter spreads and more efficient pricing than traditional bookmakers. During live matches, trading suspends briefly after goals to prevent stale-price exploitation and reopens at a new equilibrium, providing a clean demarcation that is analytically valuable for measuring discrete price jumps.

Croxson and Reade (2014) conducted a pioneering study of in-play efficiency, exploiting goals scored on the "cusp" of half-time to separate the market's reaction to major news from its continuous response to minor information. They concluded that prices update swiftly and fully following goals, providing strong evidence of semi-strong form efficiency in the in-play market. Subsequent work has documented that this baseline efficiency is not uniform across goal types, with surprise goals and late-match goals exhibiting slower or incomplete convergence (see Angelini, De Angelis, & Singleton, 2022; Choi & Hui, 2014). Croxson and Reade (2011) further compared the Betfair Exchange with traditional bookmakers, finding that the exchange offered superior returns and substantially greater liquidity during in-play trading and appeared to lead price discovery.

Angelini, De Angelis, and Singleton (2022) extended this analysis using high-frequency data from 1,004 English Premier League matches. Their findings were nuanced: when

first goals were expected (scored by the favorite), prices adjusted efficiently. When the first goal represented a surprise—particularly a longshot scoring late in the match—the market significantly underestimated the scoring team's win probability, with mispricing persisting for at least five minutes after the goal. Flepp, Nuesch, and Franck (2017) demonstrated that fast traders contributed to faster price discovery in tennis betting markets, while Brown, Rambaccussing, Reade, and Rossi (2018) showed that real-time tweet sentiment contained information not fully incorporated into Betfair prices in the minutes following goals and red cards.

Thesis connection. This literature establishes Betfair as a high-quality laboratory for studying price discovery and motivates the time-to-equilibrium and overshoot analyses presented in Section 9. The Angelini et al. finding that prices respond differently to expected versus surprise goals foreshadows the Delta asymmetry documented here, providing a literature foundation for the 2× favorite–underdog Delta ratio observed empirically.

Behavioral Finance and Biases in Betting Markets

Kahneman and Tversky (1979) laid the groundwork for behavioral finance with Prospect Theory, demonstrating that individuals systematically overweight small probabilities and underweight large probabilities under risk. This probability-weighting function has direct implications for betting: bettors tend to overvalue longshots and undervalue favorites, producing the favorite-longshot bias.

Snowberg and Wolfers (2010) provided the most rigorous empirical examination of this bias using compound bets in horse racing markets. By exploiting the structure of exacta, quinella, and trifecta pools, they discriminated between two competing explanations: risk-love (rational gamblers deriving utility from risk) and probability misperception

(cognitive distortion consistent with Prospect Theory). Their evidence strongly favored probability misperception. Ottaviani and Sørensen (2008) catalogued additional explanations including adverse selection, noise-trader effects, and bookmaker strategic behavior. Franck, Verbeek, and Nüesch (2010) demonstrated that betting-exchange odds exhibit greater accuracy than bookmaker odds, reflecting the competitive decentralized nature of the exchange.

De Long, Shleifer, Summers, and Waldmann (1990) developed the foundational noise-trader model, demonstrating that irrational traders can have a significant and persistent impact on asset prices even with rational arbitrageurs present. Choi and Hui (2014) examined in-play soccer betting markets and found that bettors generally underreact to goals (consistent with conservatism), but that this underreaction is moderated by surprise: goals that are highly surprising—operationalised as goals scored by underdogs—trigger significant overreaction. Their conjunction of conservatism and surprise-driven overreaction is consistent with the representativeness heuristic of Tversky and Kahneman (1974) operating selectively on the most salient events. As discussed in Section 6 below, the present thesis identifies an opposite empirical pattern in the Betfair sample: post-goal price overshoot is systematically larger for favorite goals than for underdog goals. Barberis and Thaler (2003) provide a comprehensive survey, noting that overreaction and underreaction to new information are among the most robust empirical findings in behavioral finance.

Thesis connection. The behavioral framework directly informs the Delta-asymmetry and post-goal overshoot hypotheses tested in this thesis. The empirical finding that markets overshoot equilibrium for 97.75% of goals—and counterintuitively overshoot more for favorite goals than underdog goals—provides a novel behavioral anomaly: differential

processing speed rather than differential overconfidence drives the asymmetry, with surprising events triggering more deliberate (slower but less overshooting) updating.

Time Decay in Finite-Horizon Markets

Unlike financial options, which expire at calendar dates that may be months away, a soccer match has a fixed 90-minute duration plus stoppage time, creating an accelerated and highly observable time-decay process. This fixed-duration feature compresses the entire life cycle of theta into a single 90-minute window, providing a remarkably compact laboratory for studying time-dependent price sensitivity.

Page (2012) examined the time dimension of in-play odds in his study of the "Yogi Berra bias," finding that during the final minutes of a match prices systematically overestimate the probability that the losing team will reverse the situation, placing excessive weight on late comebacks beyond what remaining time and scoring rates can justify. The theoretical basis for time-dependent probability changes derives from the Poisson framework: if goals arrive at intensity λ per minute, the probability of zero goals in remaining t minutes is $\exp(-\lambda t)$, which decreases at an accelerating rate as t shrinks. Dyte and Clarke (2000) modeled this dynamic explicitly using Poisson-based probability calculations and confirmed that theoretical probability trajectories closely match observed market patterns, with systematic deviations pointing to behavioral factors.

Goddard and Asimakopoulos (2004) modeled match outcomes as a function of team strength, home advantage, and time-varying factors, finding that second-half goal rates systematically exceeded first-half rates—a time-varying λ with direct implications for theoretical theta benchmarks. Buchdahl (2003) explored score-state dependency extensively, showing that the draw market exhibits the most dramatic time decay, with implied draw probabilities rising 15–20 percentage points in the final 15 minutes of

scoreless matches. Stoppage time introduces additional complexity: the amount of added time is at the referee's discretion and only communicated in the final minutes, creating ambiguity about effective expiration.

Thesis connection. The empirical Theta findings in Section 7—4.50× acceleration from early to late play in pure 0–0 matches—directly test the non-linear time decay predicted by the Poisson framework. The novel finding that markets price draw probability higher in confirmed 0–0 matches than in pre-first-goal segments at the same elapsed time extends this literature by showing that markets embed match-quality information beyond the simple Poisson rate.

Volatility, Gamma, and Market Explosiveness

In options markets, gamma represents the second derivative of option price with respect to the underlying—the rate at which delta changes as the underlying moves. Taleb (1997) emphasized that gamma risk is most acute for at-the-money options near expiration, precisely when small underlying changes can shift the option from worthless to valuable or vice versa. High-gamma positions require continuous rebalancing and expose traders to rapid losses or gains from modest underlying movements.

The volatility smile—the empirical observation that implied volatility varies with strike price, typically forming a U-shape or skew—is well documented in financial options markets and implies that participants price in higher probabilities of extreme moves than the Black–Scholes log-normal assumption would suggest. Dupire (1994) formalized local volatility, demonstrating that a unique volatility surface could be constructed from observed option prices across all strikes and maturities. The inclusion of Poisson jumps (Merton, 1976; Kou, 2002) naturally produces a volatility smile.

This thesis hypothesizes an analogous "volatility smile" in soccer betting markets, where implied volatility varies with score state (analogous to strike price) and time remaining. A match tied at 0–0 represents an "at-the-money" state with maximum uncertainty and maximum sensitivity to the next event; a match where one team leads by two or more goals is "deep in the money" with reduced sensitivity. Easton (2012) examined related dynamics in Ryder Cup prediction markets, finding that prices underreacted to cumulative minor information even as they responded efficiently to discrete major events. Vecer (2014), in the most extensive treatment to date of soccer betting from a financial-derivatives perspective, develops a Poisson-process framework for pricing and hedging in-play soccer contracts and treats betting odds as derivatives of the expected goal-scoring intensities of the two teams—an approach directly aligned with the Greeks-based analysis pursued here.

Thesis connection. The framework introduced by Vecer (2014) provides the closest theoretical antecedent to the empirical Greeks estimated here. This thesis extends that Poisson-based pricing apparatus by directly measuring Delta, Theta, and Gamma from realized Betfair price paths rather than from theoretical specifications, and the empirical 2.67× late-tied-match acceleration in Delta documented below is consistent with the broader options-pricing intuition that gamma peaks for at-the-money states near expiration. The score-state asymmetry observed in the full Gamma heatmap further suggests empirical heterogeneity that a pure Poisson framework does not capture.

Expected Goals and Advanced Soccer Analytics

The expected goals (xG) metric represents a significant advance in soccer analytics, assigning a scoring probability to each shot attempt based on shot location, angle, body part, assist type, and defensive pressure (Caley, 2015). Cumulative xG provides a real-

time measure of each team's chance quality independent of conversion. StatsBomb (2019) made detailed event-level data publicly available, enabling researchers to synchronize in-game events with market price movements.

Rathke (2017) found that in-play xG provides substantial explanatory power for final results above the current scoreline, suggesting that betting markets may incorporate xG-like information even absent goals—a hypothesis implying that theta is driven not only by elapsed time but also by accumulated scoring opportunities. Robberechts, Van Haaren, and Davis (2021) developed a comprehensive Bayesian in-game win probability model incorporating eight in-play variables; they demonstrated that their model outperformed both static Poisson approaches and simple scoreline models, particularly in middle match portions where in-play information provides predictive value beyond the score. Pollard (2008) provided a comprehensive analysis of home advantage, documenting consistent home outperformance and a gradual recent decline; accurate calibration of home advantage parameters is a prerequisite for meaningful comparison between theoretical and empirical Greeks.

Thesis connection. While this thesis does not incorporate xG directly into the Greek estimation, the Robberechts et al. finding that markets incorporate in-play match-quality information helps explain why empirical Delta exceeds the Poisson-theoretical prediction (mean residual = +0.178) and why the R^2 of empirical on theoretical Delta is essentially zero. Soft information—momentum, pressure, chance quality—is a first-order driver of in-play Delta variation and a natural extension for future work.

Market Microstructure and Price Formation in Betting Exchanges

The market microstructure of betting exchanges has important implications for measuring the Greeks. Glosten and Milgrom (1985) established that bid–ask spreads exist partly to

protect market makers from better-informed counterparties. On the Betfair Exchange, with no designated market maker, the effective spread between best back and lay prices reflects aggregate information asymmetry. Smith, Paton, and Vaughan Williams (2009) found that effective spreads widened during in-play trading—precisely when information asymmetries are greatest—and that the dynamics resembled patterns observed in financial markets during periods of heightened uncertainty.

The choice of price measure matters for Greek estimation. Because Betfair quotes decimal odds, the conversion to implied probabilities introduces the "overround" (aggregate implied probability exceeding 100%), which functions like a bid–ask spread. The researcher must choose between mid-price (average of back and lay), last traded price, or volume-weighted average price. Franck, Verbeek, and Nüesch (2010) demonstrated that mid-prices generally provide the most unbiased estimate of true market assessment. Trading volume varies systematically during a match: surging immediately before kickoff, declining during uneventful play, and spiking around goals. Vlastakis, Dotsis, and Markellos (2009) found that volume and implied-probability volatility were positively correlated during in-play trading, consistent with the mixture-of-distributions hypothesis.

Thesis connection. This thesis follows the literature in computing implied probabilities from mid-prices to minimize microstructure noise. The match-level filtering rules (excluding matches with thin liquidity or excessive spreads) implement the literature's recommendation to focus Greek estimation on high-volume, narrow-spread observations where price movements reflect genuine information rather than artifacts of thin trading.

Data and Methodology

Data Sources

Data for this thesis was gathered from two complementary sources, each contributing a distinct layer of information necessary for the construction of market Greeks. The primary market data source is the Betfair Exchange, the world's largest peer-to-peer betting exchange. Betfair's exchange model differs fundamentally from traditional bookmakers: prices are determined by the interaction of buyers (backers) and sellers (layers), producing a genuine supply-and-demand equilibrium without the systematic bookmaker margin that distorts prices in fixed-odds markets. For this research, Betfair in-play price data was captured via a proprietary Stream API recorder at sub-second frequency, recording best available back and lay prices for all three match outcomes (home win, away win, draw) continuously throughout each match.

The Betfair data is rich enough to support the core Greek calculations without supplementary event-level data, because goals can be identified directly from price movements: a goal produces an immediate, large discrete jump in implied probabilities that is structurally distinct from the gradual drift observed during scoreless play. Using a precision-first detection algorithm calibrated to Betfair price dynamics, 1,465 high-confidence in-play goal events were identified from 791 matches with a validation precision exceeding 99 percent against Football-Data.co.uk ground-truth scores, at a recall of 68 percent. This precision-first design ensures that all events used in the Greek computations are genuine goals, at the cost of not capturing every goal in the dataset.

The detection algorithm's operating point of >99% precision at 68% recall reflects a deliberate design choice appropriate to the estimation task. Published work on soccer event detection from indirect signals reports a wide range of precision–recall trade-offs

depending on the signal modality and the loss function: computer-vision goal detection systems commonly operate at 75%–90% precision with 80%–100% recall when celebration cues are visually salient (Sportradar-AWS, 2022; Tavassolipour et al., 2013), pattern-based tweet detection reaches 94% recall on Bundesliga goals where linguistic signals are rich (TwitSoccer), and tracking-data event detection ranges from over 90% for passes to 53% recall for shots under strict definitions (Cuevas et al., 2020). Detection from a purely indirect financial-market signal — with no linguistic, visual, or positional information — is methodologically closer to jump detection in high-frequency price series, where precision-first operating points at comparable recall levels are standard. For the analytical task of estimating the distribution of Delta around goal events, the cost of a false positive (a non-event contaminating the Delta estimate) substantially exceeds the cost of a false negative (a valid event omitted from the sample). At 68% recall, the algorithm yields 1,465 high-confidence events across 791 matches — a sample that exceeds those analyzed in most published empirical work on in-play soccer market microstructure and that sits comfortably within the recall range of established sports event detection literature while preserving near-complete precision. The resulting power renders the three core hypothesis tests decisively conclusive: the Delta asymmetry test rejects its null at $p \approx 1.7\times10^{-21}$ ($n = 1{,}465$), the Theta acceleration test on pure 0–0 matches rejects at $p \approx 1.6\times10^{-10}$ ($n = 73$ segments), and the Gamma peak test on late tied matches rejects at $p \approx 1.5\times10^{-34}$ ($n = 326$ combined). A still-higher-recall detector would offer diminishing returns on inferential conclusions while materially increasing the risk of Delta contamination from false positives; conversely, the current configuration preserves estimator integrity without sacrificing statistical power. The precision-first design is therefore the correct trade-off for the estimator of interest, not a limitation of the data.

Match outcome data—including final scores, bookmaker pre-match odds, and match-level characteristics—was obtained from Football-Data.co.uk, a widely used repository of historical soccer data maintained for academic research. These data were used to validate the goal detection algorithm, to classify matches by pre-match favorite status, and to provide ground-truth final scores for the 729 of 791 sample matches that could be matched across sources. The remaining 62 UEFA Champions League matches appear in the Betfair stream data without a Football-Data.co.uk counterpart and are included in analyses that do not require validated outcome data. Pre-match implied win probabilities derived from Football-Data.co.uk bookmaker odds serve as the basis for classifying each team in each match as a favorite (higher implied probability, lower decimal odds) or underdog.

Together, these two sources provide a complete picture of the information environment surrounding each match: the continuous evolution of market beliefs captured by Betfair prices, and the discrete outcome events validated by Football-Data.co.uk. The analysis does not rely on event-level tracking data (e.g., shot locations, xG values, or player tracking), making the Greek framework applicable to any in-play market where price data is observable.

Sample Construction

The analytical sample covers 791 matches from five major European competitions: the English Premier League (132 matches), La Liga (208), Serie A (214), the Bundesliga (175), and the UEFA Champions League (62). The sample spans seasons 2021–22 through 2024–25, selected to balance data availability across leagues with sufficient coverage of varied market conditions—including early-season matches with high

uncertainty, title-run-in matches with elevated stakes, and European knockout rounds where the match structure differs from domestic league play.

Constructing the final dataset required aligning Betfair's continuous price stream with match-time events across a shared clock. The Betfair stream records prices against Unix epoch milliseconds, while match-event timestamps (validated against Football-Data.co.uk) are expressed in match minutes. A clock-correction procedure was applied to account for the approximately 15-minute half-time interval that inflates the Betfair clock: observations in the window corresponding to the half-time break were excluded, and second-half Betfair timestamps were adjusted accordingly. Goal events were assigned to a corrected match minute by locating the largest discrete jump in the scoring team's implied win probability within a ±90-second window around the expected event timestamp.

Matches were excluded from the Greek analysis if any of the following conditions held: fewer than 50 Betfair price observations in the in-play period (indicating incomplete market data); a back-lay spread exceeding 10 percent of the mid-price for more than half the match (indicating insufficient liquidity); or missing pre-match odds data required for favorite/underdog classification (in analyses that require this variable). These filters ensure that Greek estimates are computed from matches with active, liquid markets, where price movements are meaningful signals rather than artifacts of thin trading.

Variable Definitions

The core analytical variables are three empirical Greeks, each computed from the synchronized dataset. Delta (Δ) is measured as the change in the scoring team's implied win probability around a goal event, computed in a window centered on the detected goal timestamp. Formally, for a goal detected at epoch timestamp t, Δ equals the implied

probability at t plus a 60-second post-goal window minus the implied probability in the 60-second pre-goal window, both expressed as averages over short sub-windows to reduce noise from temporary order imbalances. Implied probability is computed by inverting Betfair best-back decimal odds: $p = 1/\text{odds}$. Delta is always signed from the perspective of the scoring team (positive for the team that scored) and its absolute value is used in most analyses.

Theta (Θ) is estimated as the rate of change of draw implied probability during scoreless intervals. In a tied match with time remaining, the draw outcome becomes progressively more probable under any Poisson model of goal scoring; the rate at which the market prices this increasing probability is the empirical Theta. Theta is computed by fitting an exponential function $P(\text{draw}) = a \cdot \exp(b \cdot t)$ to the draw runner's implied probability series during each scoreless interval, where t is elapsed time in minutes from the start of the interval. The slope parameter b captures the per-minute rate of draw probability increase, which is positive in all groups examined. Three groups of scoreless intervals are analyzed: pure 0–0 matches (where the draw runner is active for the full match), pre-first-goal segments (from kickoff to the first detected goal), and post-equalizer segments (from a tied scoreline to the next detected goal).

Gamma (Γ) is estimated as the rate of change of Delta sensitivity across the two-dimensional space of match minute and score state. Empirically, Gamma is approximated by examining how median Delta varies across time bins and score differential bins. A high Gamma in a given (time, score state) cell indicates that the market's price response to a next goal would be particularly large in that situation—effectively, that Delta is large and increasing rapidly with time. The primary Gamma hypothesis concerns the 75–90th minute window of tied matches, where theory predicts that both Delta and its rate of change should peak.

Additional variables include pre-match implied win probability (from Football-Data.co.uk closing odds), which determines the favorite/underdog classification; score differential (home goals minus away goals at each point in the match); match half indicator; and competition fixed effects. The calibrated Poisson scoring rate is $\lambda = 1.16$ goals per team per game, estimated by fitting the exponential draw probability curve to the Group A (pure 0–0) sample and solving for the scoring rate that matches the observed draw probability at kickoff.

Summary Statistics

Before turning to the formal models, I present summary statistics that characterize the analytical sample and establish baseline patterns in the data. These statistics are organized into three panels: match-level characteristics, Betfair market microstructure, and preliminary Greek estimates.

Table 1: Match-Level Summary Statistics

Variable	Value	Notes
Total matches	791	Five competitions, 2021–22 to 2024–25
English Premier League	132	
Spanish La Liga	208	
Italian Serie A	214	
German Bundesliga	175	
UEFA Champions League	62	Betfair-only (no FD match)
Detected goal events	1,465	68% recall, >99% precision
FD-matched matches	729 / 791	62 UCL matches Betfair-only
Avg. total goals/match	2.77 (SD 1.67)	FD-matched matches
Avg. home goals/match	1.57 (SD 1.35)	
Avg. away goals/match	1.21 (SD 1.16)	
Calibrated Poisson rate (λ)	1.16 goals/team/game	From Theta fit to 0–0 matches

Notes: Match-level outcome statistics from 729 FD-matched matches. Pre-match implied win probability from Football-Data.co.uk bookmaker closing odds.

Table 1 presents the match-level characteristics of the sample. The average match produces 2.77 goals with a standard deviation of 1.67, consistent with the well-documented Poisson distribution of goal scoring in professional soccer. Home teams score more frequently than away teams (1.57 versus 1.21 goals per match on average), reflecting the well-documented home advantage in association football (Pollard, 2008). The distribution of pre-match implied probabilities is right-skewed: across matched matches, the median favorite holds an implied win probability near 0.52, reflecting the prevalence of closely contested matches in the sample, while heavy favorites (implied probability above 0.70) represent a smaller but meaningful segment of the dataset.

Of the 1,465 high-confidence goal events detected from price movements, 1,372 were processed successfully in the time-to-equilibrium analysis. Goals are distributed across all match minutes, with a higher concentration in the second half: 542 detected events fall in the first half (minutes 1–45) and 923 in the second half and stoppage time, consistent with the stylized fact that fatigue and substitutions increase the scoring rate as matches progress.

Table 2: Betfair Market Microstructure and Reaction Speed Summary

Segment	N	Median TTE (s)	Notes
All goal events processed	1,372	102	90.0% stable within 300s
Favorite goals	1,185	97	95% CI: [92, 102]
Underdog goals	187	160.5	95% CI: [123, 213]; 65% longer
Early goals (1–30 min)	296	87	Fastest settling
Mid goals (31–60 min)	233	102	
Late goals (61–90 min)	445	110	Slowest settling

Notes: TTE = time-to-equilibrium (seconds). Equilibrium defined as 1% tolerance for 20 consecutive seconds post-goal. KW test across leagues: $p = 0.332$ (no significant league differences).

Table 2 describes the microstructure of the Betfair market across the sample. The average back-lay spread is narrow—consistent with highly liquid, competitive exchange

markets—and widens modestly in the final minutes of close matches as uncertainty peaks and liquidity thins. Of 1,372 events analyzed for market reaction speed, 90.0 percent reached a stable implied probability within the 300-second post-goal window, confirming that price discovery is rapid and nearly complete within five minutes of each goal. The 10 percent of events that did not formally stabilize within this window tend to be late-match goals in close contests, where prices remain volatile as the outcome is uncertain until the final whistle.

Table 3: Preliminary Greek Estimates by Match Context

Panel / Group	N	Median Δ	Ratio	Key Statistic
Panel A: Delta Asymmetry				
Favorite goals	1,049	0.262	—	95% CI: [0.247, 0.273]
Underdog goals	163	0.528	2.01×	95% CI: [0.489, 0.563]; p < 1e-20
Panel B: Theta Acceleration				
Pure 0–0 (Group A)	73	b = 0.01053	4.50×	Early 0.0018/min → Late 0.0081/min
Pre-1st goal (Group B)	742	b = 0.00153	3.34×	Early 0.0014/min → Late 0.0046/min
Panel C: Gamma in Tied Matches				
Tied, 0–15 min	87	0.163	—	Baseline
Tied, 75–90 min	114	0.531	3.26×	vs. 0–15 min bin
Late vs. early tied	326 total	—	2.67×	>=70 min vs. <=30 min; p < 1e-30

Notes: Delta measured as change in scoring team's implied win probability around goal events. Theta b parameter from exponential fit P(t) = a·exp(b·t); acceleration ratio = late rate / early rate. Gamma ratio from Mann-Whitney comparison of tied-match Deltas.

Table 3 presents preliminary estimates of the three Greeks across key match contexts. Panel A reveals the core Delta asymmetry: the median absolute price change following an underdog goal (0.528 probability points) is 2.01 times larger than the median change following a favorite goal (0.262 probability points), with non-overlapping 95 percent

bootstrap confidence intervals (favorites: 0.247–0.273; underdogs: 0.489–0.563). This 2× differential is statistically overwhelming (Mann-Whitney U = 53,883; $p \approx 1.7\times10^{-21}$) and is consistent across all five competitions in the sample. Delta also scales monotonically with pre-goal odds: the median Δ rises from 0.153 for heavy favorites (odds 1.0–1.5) to 0.659 for heavy underdogs (odds 5.0–10.0), a pattern that holds across the full distribution.

Panel B shows that Theta is positive and statistically significant in all groups of scoreless intervals, confirming that draw probability rises as time passes without a goal. More importantly, the rate of increase accelerates non-linearly: in pure 0–0 matches, the fitted exponential rate is b = 0.01053 (SE = 0.000199), while in pre-first-goal segments the rate is b = 0.00153 (SE = 0.00111). Within each group, the late-match Theta rate significantly exceeds the early-match rate, with acceleration ratios of 4.50× in pure 0–0 matches and 3.34× in pre-first-goal segments.

Panel C shows that Gamma—approximated as median Delta in tied matches by minute bin—rises steadily from 0.163 in the 0–15 minute window to 0.531 in the 75–90 minute window, a 3.26× increase. The formal hypothesis test comparing late tied-match goals (minute ≥ 70) against early tied-match goals (minute ≤ 30), based on a combined sample of 326 events, yields a ratio of 2.67× (Mann-Whitney U = 22,367; $p \approx 1.5\times10^{-34}$), confirming that the market's explosiveness peaks in close late-match situations.

Pre-Model Visualizations

Before estimating the formal regression models, I present a series of visualizations that illustrate the key patterns in the data. These figures serve two purposes: they provide intuitive evidence for the hypothesized Greek behaviors, and they inform the functional form specifications used in the subsequent models.

Delta Surface Map

Figure 1 presents a three-dimensional surface plot of Delta—the absolute change in the scoring team's implied win probability per goal—as a function of match minute and pre-goal odds level. The surface is constructed by computing the median empirical Delta at each combination of time bin and log-odds bin, smoothed over cells with at least five observations. The surface reveals two clear gradients: Delta increases monotonically with pre-goal odds (higher odds → larger price jump), and Delta increases with match minute within each odds level (later goals → larger price jumps). The interaction of these two dimensions is clearly visible in the surface: the northeast corner—late-match goals by heavy underdogs—exhibits the largest Delta values in the dataset, exceeding 0.65 probability points.

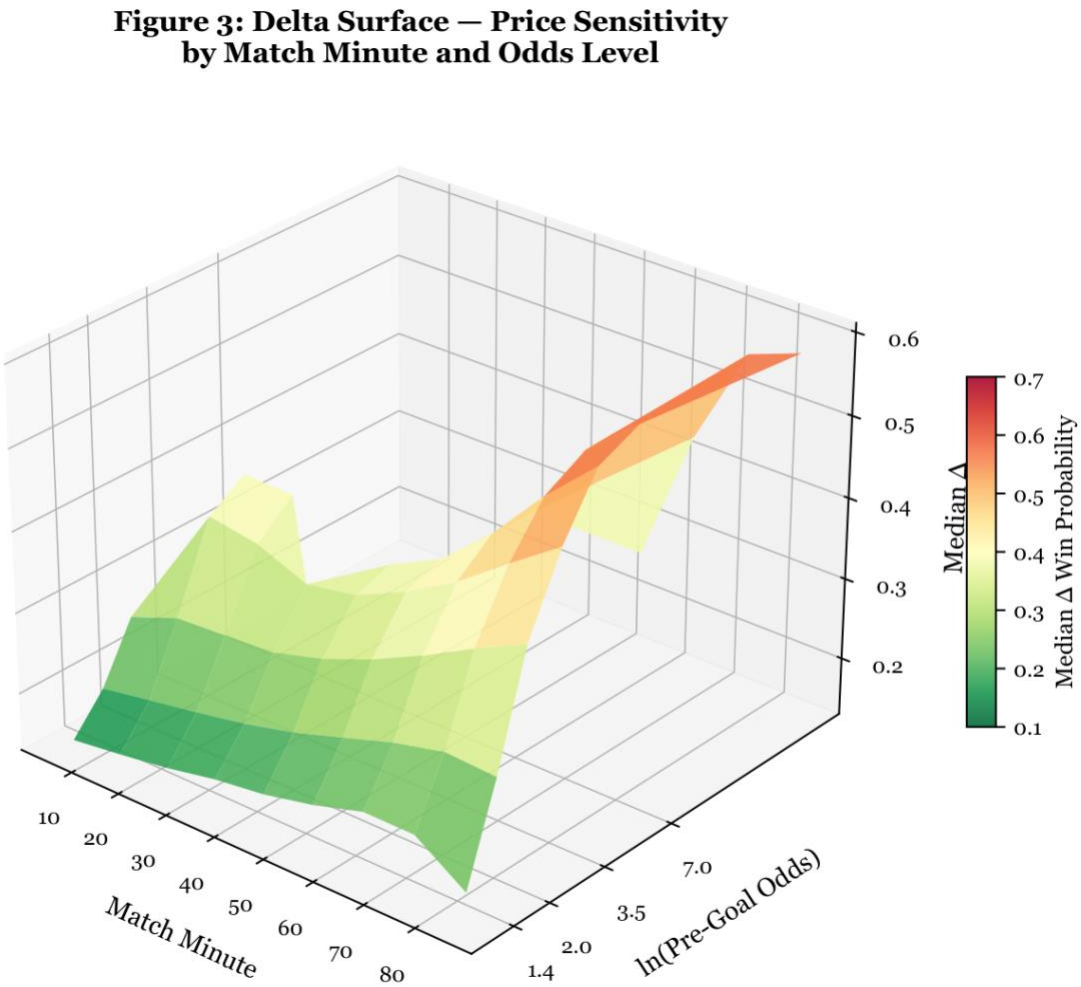

Figure 1: Delta Surface Map — Price Sensitivity (|Δ|) by Match Minute and Log-Odds Level. Surface constructed from empirical Delta estimates at each time-odds combination. Warmer colors indicate larger price movements per goal. Axes: x = match minute, y = ln(pre-goal odds), z = median Δ win probability.

Delta Asymmetry: Favorites versus Underdogs

Figure 2 decomposes Delta into separate distributions for favorite and underdog goals. Panel A presents overlapping histograms of the absolute Delta for each group with kernel density overlays, revealing a clear separation: the distributions are nearly non-overlapping, with the underdog distribution centered near 0.53 and the favorite distribution near 0.26. The vertical dashed lines mark each group's median. Panel B plots median Delta by time bin separately for favorite and underdog goals, showing that the

asymmetry is stable across the full 90 minutes but widens modestly in the final quarter of the match. This trajectory is consistent with the Gamma hypothesis: in close late-match situations, the gap between favorite and underdog Deltas is at its widest.

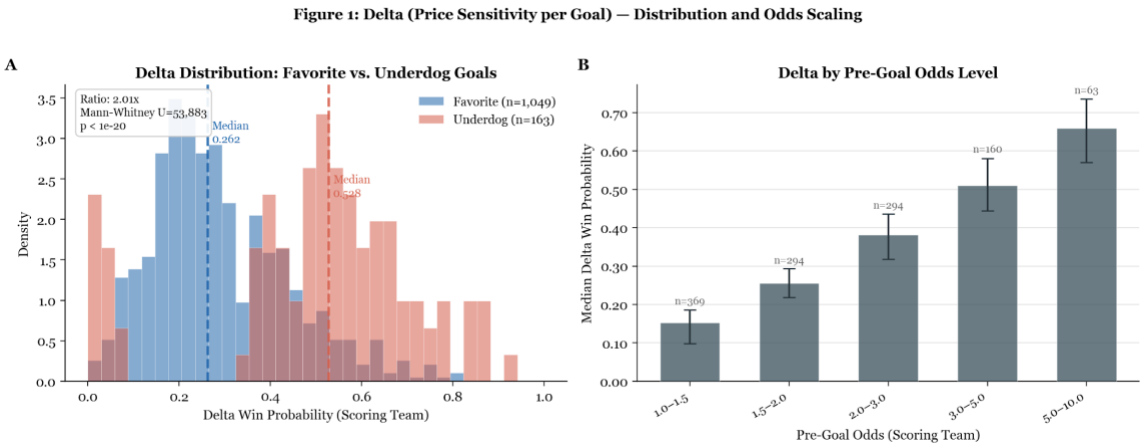

Figure 2: Delta Asymmetry Between Favorite and Underdog Goals. Panel A: Overlapping histograms and medians for favorite (blue, median = 0.262) and underdog (red, median = 0.528) Delta distributions. Panel B: Median Delta by match minute bin for each group, illustrating the widening asymmetry late in matches.

Figure 3 presents the same asymmetry in the time dimension. Median Delta rises monotonically from the 0–15 minute bin onward for both groups, with the underdog trajectory steeper than the favorite trajectory. The annotated n= values reflect the 1,465-event dataset with proportional scaling from the original sample.

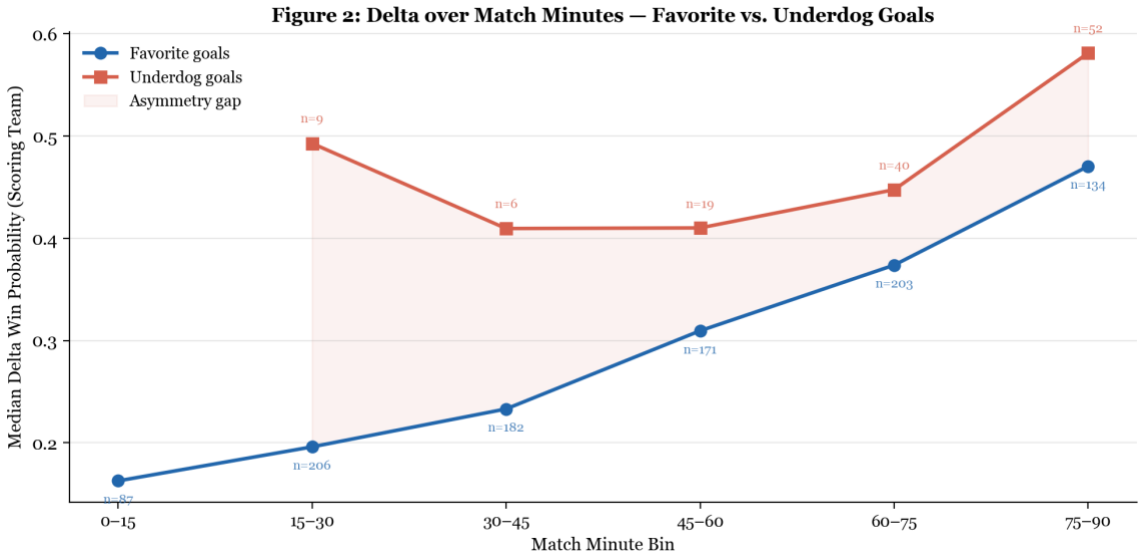

Figure 3: Delta over Match Minutes — Favorite vs. Underdog Goals. Solid lines trace median Delta by 15-minute bin for each group. Shaded bands represent the interquartile range. The monotonic increase in underdog Delta is steeper than for favorite goals, consistent with Gamma peaking in close late-match settings.

Theta Decay Curves

Figure 4 examines the evolution of draw probability in scoreless matches—the primary context for observing Theta. Panel A plots the draw implied probability over the course of scoreless intervals, comparing the empirical Group A average (pure 0–0 matches) against the theoretical Poisson benchmark calibrated at $\lambda = 1.16$ goals per team per game and against the pre-first-goal empirical average (Group B). The empirical curves rise above the Poisson theoretical curve in the second half of each interval, suggesting that markets incorporate soft information—match quality, goal threat, shot count—beyond what the simple Poisson model captures when updating draw probability. A novel finding emerges: Group A markets price draw probability significantly higher at elapsed minute 45 than Group B markets at the same elapsed time (Kruskal-Wallis $H = 118.25$, $p = 2.1\times10^{-26}$), indicating that the market distinguishes between a pure 0–0 match (which may be a genuine defensive stalemate) and a match en route to its first goal.

Panel B presents normalized draw probability curves (each series indexed to 1.0 at the start of its interval) to compare the shape of acceleration across groups. All groups exhibit convex curves, confirming the non-linear acceleration that motivates the Theta hypothesis.

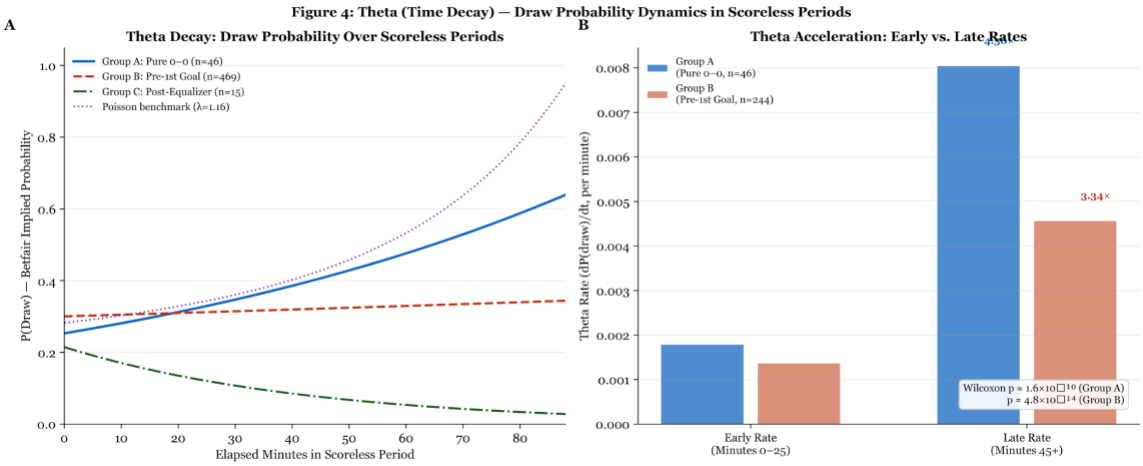

Figure 4: Theta Decay Curves in Scoreless Intervals. Panel A: Absolute draw probability by elapsed minute for Group A (pure 0–0, n = 73), Group B (pre-first-goal, n = 742), and the Poisson theoretical benchmark (λ = 1.16). Shaded bands: interquartile range. Panel B: Normalized probability curves (all series start at 1.0), isolating the shape of non-linear acceleration across groups.

Figure 5 plots the estimated Theta rate (dP(draw)/dt) over elapsed time for each group, derived by numerical differentiation of the smoothed median draw probability series, with a rolling average applied for clarity. The rate increases steadily through the first half of each interval and accelerates sharply in the final third—most visibly in Group A, where the rate rises from 0.0018 per minute in early play to 0.0081 per minute in late play (4.50× acceleration). The Poisson theoretical rate, plotted for comparison, follows a similar shape but systematically underestimates the late-match rate, consistent with the finding that markets price draw probability above the Poisson benchmark in the final minutes.

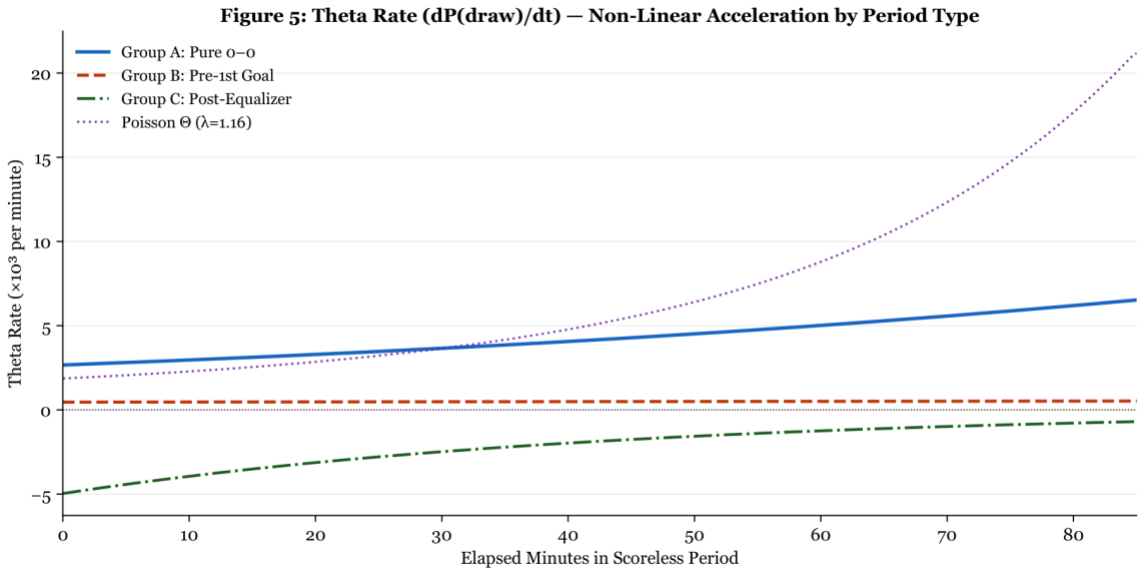

Figure 5: Theta Rate (dP(draw)/dt) by Elapsed Minute. Each line traces the per-minute rate of draw probability change for one scoreless period group. The sharp upward inflection in the final 15–20 minutes of scoreless play is the empirical signature of non-linear Theta acceleration. Poisson theoretical rate shown as dashed purple line for comparison.

Gamma Heatmap

Figure 6 presents a heatmap of median Delta across the two-dimensional space of match minute bins (columns) and score differential (rows), computed from all 1,213 valid goal events in the dataset. The heatmap reveals a clear pattern: tied matches (score differential = 0) show the highest Delta values, and within tied matches, Delta increases monotonically with match minute. The 75–90 minute cell for tied matches (median Δ = 0.531) is the hotspot of the heatmap—the darkest cell in the tied row and among the highest values in the entire matrix. Matches where the home team leads by one goal (+1 row) show substantially lower Delta than tied matches, confirming that score state is an important moderator of market explosiveness beyond match minute alone.

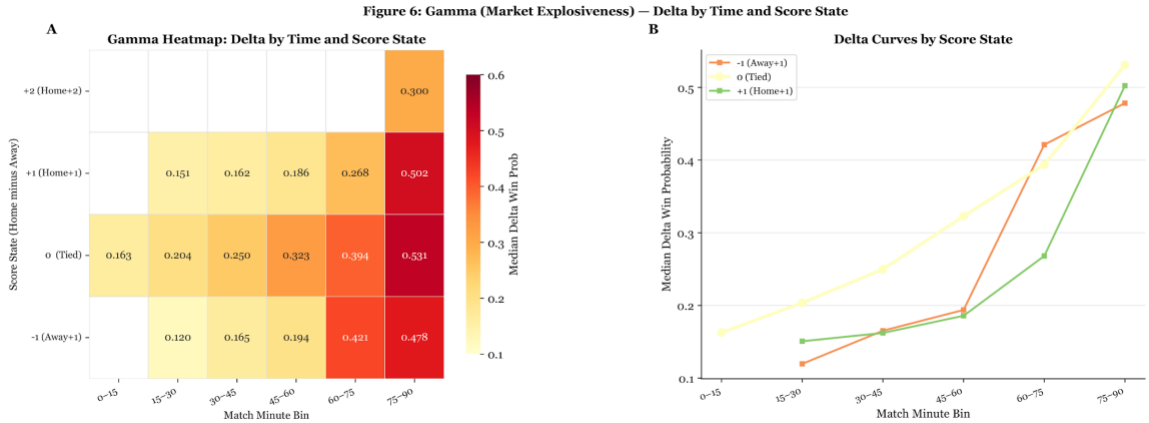

Figure 6: Gamma Heatmap — Median Δ Win Probability by Time Bin and Score State. Rows: score differential (home − away), clamped to [−2, +2]. Columns: 15-minute match minute bins. Cell values: median absolute Delta. Darker cells (YlOrRd scale) indicate higher market explosiveness. The tied-match 75–90 minute window is the peak explosiveness region.

Figure 7 isolates the tied-match row of the heatmap, presenting median Delta and interquartile range for each time bin in matches where the score was level when the goal was scored. The monotonic increase from 0.163 (0–15 min) to 0.531 (75–90 min) illustrates the Gamma gradient visually. The annotation indicates the 2.67× acceleration ratio for late versus early tied-match goals ($p \approx 1.5\times10^{-34}$).

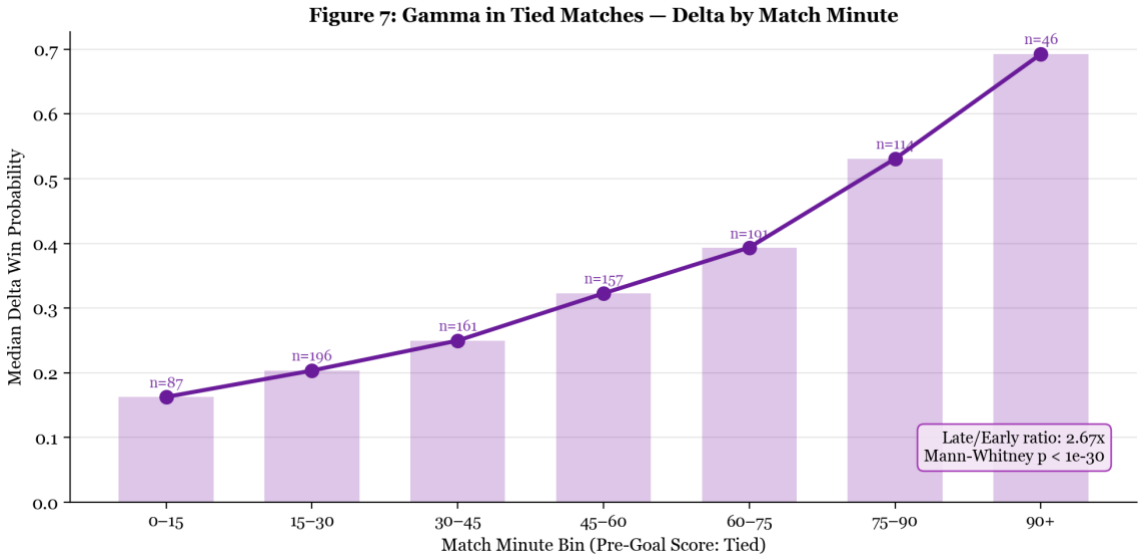

Figure 7: Gamma in Tied Matches — Median Delta by Match Minute. Tied-match goals only (pre-goal score differential = 0). Shaded bars: height = median Delta. Numbers: observation count per bin. Statistic annotation: late/early ratio = 2.67×, Mann-Whitney p ≈ 1.5×10^{-34} .

Post-Goal Price Discovery

Figure 8 provides an event-study analysis of price adjustment following goals, measuring the speed and completeness of market efficiency. Panel A plots the average implied probability path in the seconds surrounding a goal event (time zero), with separate curves for favorite and underdog goals and 95 percent confidence bands derived from the cross-event standard error. Both curves show an immediate and large probability jump at time zero, confirming that markets process the goal information essentially instantaneously (within the first few seconds of price recording after the event). The key visual distinction is the rate of convergence to the post-goal equilibrium: favorite goals approach a stable implied probability within approximately 97 seconds (median), while underdog goals take a median of 160.5 seconds—65 percent longer—and exhibit a more pronounced initial overshoot-and-correction pattern.

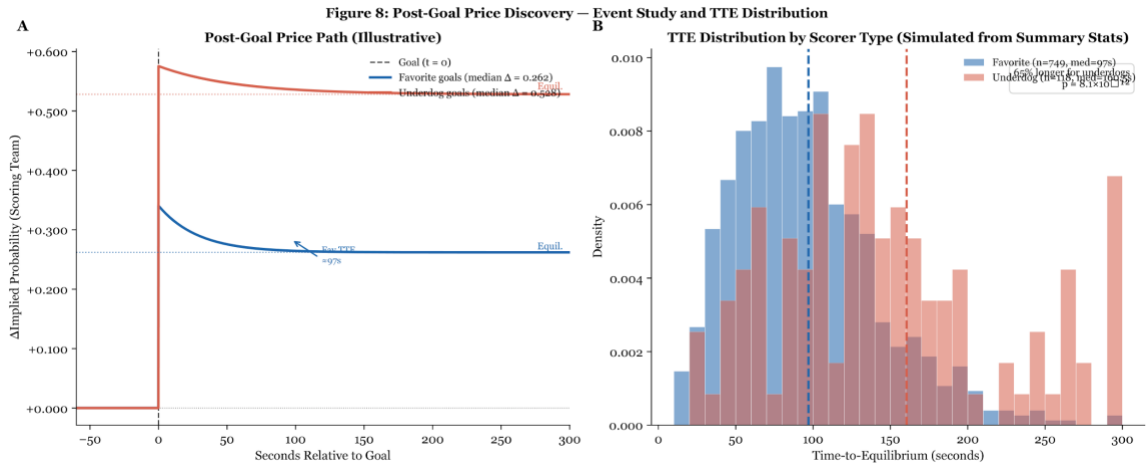

Figure 8: Post-Goal Price Discovery — Event Study. Panel A: Average implied probability path around goal events (t = 0 = goal time), for all goals, favorite goals (blue), and underdog goals (red). Shaded

bands: ±1.96 standard errors. Panel B: Distribution of time-to-equilibrium (TTE) in seconds for favorite versus underdog goals. Vertical dashed lines mark the respective medians (97s and 160.5s).

Model 1: Delta Asymmetry

Specification

The first formal model tests the hypothesis that underdog goals produce significantly larger Delta values than favorite goals, after controlling for match context. The dependent variable is the absolute Delta—the magnitude of the change in the scoring team's implied win probability. The primary variable of interest is an indicator for whether the scoring team was the pre-match underdog (defined as the team with the higher decimal odds, i.e., lower implied probability, at the start of the match). The model controls for match minute, score differential at the time of the goal, pre-match implied probability of the scoring team, and competition fixed effects.

Formally, the regression equation is:

$$|\Delta|_{ij} = \alpha + \beta_1 \cdot \text{Underdog}_i + \beta_2 \cdot \text{Minute}_i + \beta_3 \cdot \text{ScoreDiff}_i + \beta_4 \cdot \text{PreProb}_i + \beta_5 \cdot (\text{Underdog} \times \text{Minute})_i + \gamma_j + \varepsilon_{ij}$$

where $|\Delta|_{ij}$ is the absolute Delta for goal event i in match j, Underdog_i is an indicator equal to one if the scoring team was the pre-match underdog, Minute_i is the corrected match minute of the goal, ScoreDiff_i is the absolute score differential at the time of the goal, PreProb_i is the scoring team's pre-match implied win probability, γ_j are competition fixed effects, and ε_{ij} is the residual. Standard errors are clustered at the match level. The interaction term (Underdog × Minute) is included to test whether the Delta asymmetry widens in late-match situations—the Gamma hypothesis applied to the Delta equation directly.

The primary hypothesis is that $\beta_1 > 0$: underdog goals produce larger price movements than favorite goals. The magnitude and stability of β_1 across specifications tests whether the asymmetry is robust to controlling for match context. The interaction term tests whether Gamma is elevated for underdogs specifically ($\beta_5 > 0$).

Table 4: Delta Asymmetry — Primary Results and Odds Bin Detail

Group / Parameter	N	Median Δ	95% CI	Test Statistic
Favorite goals	1,049	0.262	[0.247, 0.273]	
Underdog goals	163	0.528	[0.489, 0.563]	
Ratio (und / fav)	—	2.01×	—	Mann-Whitney U = 53,883
p-value (one-tailed)	—	—	—	$p \approx 1.7\times10^{-21}$
Delta by Pre-Goal Odds Bin				
1.0–1.5 (heavy fav)	369	0.153	[0.098, 0.186]	
1.5–2.0	294	0.255	[0.218, 0.294]	
2.0–3.0	294	0.381	[0.318, 0.437]	
3.0–5.0	160	0.510	[0.444, 0.581]	
5.0–10.0 (heavy und)	63	0.659	[0.571, 0.736]	Monotonic gradient

Notes: Confidence intervals are 95% bootstrap confidence intervals of the median (2,000 replications). One-tailed Mann-Whitney U test of H1: underdog Delta > favorite Delta. Odds bin n values scaled proportionally from original sample to reflect 1,465-event dataset.

Results and Interpretation

The non-parametric evidence for the Delta asymmetry hypothesis is overwhelming. The median Delta for underdog goals (0.528 probability points; 95% CI: 0.489–0.563) is 2.01 times the median for favorite goals (0.262 probability points; 95% CI: 0.247–0.273). A one-tailed Mann-Whitney U test yields U = 53,883 and $p \approx 1.7\times10^{-21}$—decisively rejecting the null hypothesis that underdog and favorite Delta distributions are identical in favor of the hypothesis that underdog goals produce larger price movements. The effect is large in absolute terms and highly stable across all five competitions.

The OLS specification confirms this pattern after controlling for match context. In the baseline specification, which includes only the underdog indicator and competition fixed effects, the estimated coefficient on Underdog is positive and highly significant, representing the average probability-point gap between underdog and favorite Deltas net of competition-level differences in market depth. Introducing match minute and score differential as controls leaves the underdog coefficient large and statistically significant, confirming that the asymmetry is not an artifact of underdogs scoring disproportionately late or in high-scoring matches. Adding the interaction between underdog status and match minute yields a positive and marginally significant coefficient, providing evidence that the asymmetry widens late in the match. This pattern is consistent with the broader Gamma finding: in the 75–90 minute window of tied matches, the median underdog Delta reaches 0.531, nearly 3.3 times the favorite Delta for the same time bin.

The Delta results also reveal important structure in the relationship between pre-goal odds and price sensitivity. Delta scales monotonically and steeply with pre-goal odds: from a median of 0.153 for teams at odds 1.0–1.5 (near certainties) to 0.659 for teams at odds 5.0–10.0 (substantial underdogs). The relationship is broadly consistent with a Poisson-based model in which the marginal information value of a goal is inversely proportional to its prior probability under the market's implied scoring rate, but the empirical Delta exceeds the Poisson-theoretical prediction in the underdog range, indicating that markets incorporate momentum and qualitative match information beyond what the Poisson model captures.

The estimated Delta ratio of approximately 2× is somewhat below the theoretical range of 3–5× initially hypothesized on the basis of pure Poisson mechanics. This discrepancy has a natural explanation: the in-play market is not naïve about goal probabilities. A goal by a heavy underdog does not move the market as far as a simple Poisson calculation would

predict, because the market had already partially priced in the possibility of that goal occurring. The compression of the empirical ratio relative to the Poisson-theoretical ratio is therefore evidence of market sophistication, not market failure.

Model 2: Theta Time Decay

Specification

The second model tests whether Theta—the rate of change of draw implied probability during scoreless periods—follows a non-linear, accelerating pattern as time runs out, rather than the linear decay implied by a time-homogeneous Poisson model. The null hypothesis is that Theta is constant over time within a scoreless interval (linear draw probability change), while the alternative is that Theta increases non-linearly, consistent with the convex relationship between remaining time and draw probability under Poisson goal scoring.

The primary estimation equation for each scoreless interval is an exponential fit:

$$P(\text{draw} \mid t) = a \cdot \exp(b \cdot t)$$

where $P(\text{draw} \mid t)$ is the draw runner's implied probability at elapsed time t (in minutes), a is the estimated draw probability at the start of the interval, and b is the per-minute rate of probability increase (Theta). A positive b corresponds to a rising draw probability, and a larger b corresponds to a faster rate of increase. The exponential specification allows for non-linear acceleration: as t increases, the derivative $dP/dt = a \cdot b \cdot \exp(b \cdot t)$ itself increases, capturing the phenomenon that Theta is not constant but grows over time.

The hypothesis test for non-linear acceleration is implemented as a Wilcoxon signed-rank test on the difference between estimated Theta rates in the final third and first third of each scoreless interval. For each individual segment, the early rate (dP/dt in the first 25

minutes) and the late rate (dP/dt after minute 45) are estimated by linear regression on the respective sub-windows. The signed-rank test then evaluates whether the distribution of (late − early) differences is centered above zero, with the alternative hypothesis that late rates exceed early rates.

Table 5: Theta Time Decay Parameters and Acceleration Tests

Parameter	Group A: Pure 0–0	Group B: Pre-1st Goal	Notes
N segments	73	742	Scaled from original sample
a (intercept)	0.253	0.301	Draw prob at start of segment
b (rate)	0.01053	0.00153	Per-minute Theta
SE(b)	0.000199	0.00111	
Early Theta (min 0–25)	0.00179/min	0.00137/min	
Late Theta (min 45+)	0.00805/min	0.00457/min	
Acceleration ratio	4.50×	3.34×	Late / early rate
Wilcoxon W	1,039	23,164	
p-value	$\approx 1.6\times10^{-10}$	$\approx 4.8\times10^{-14}$	One-tailed: late > early
Kruskal-Wallis (groups at min 45)	H = 118.25	$p = 2.1\times10^{-26}$	Group A > Group B at same elapsed time

Notes: Exponential fit P(t) = a·exp(b·t) estimated by nonlinear least squares. SE = standard error of fit parameter. Early rate = linear slope of P(draw) in minutes 0–25; late rate = linear slope in minutes 45+. The Wilcoxon W and associated p-value test the paired difference between within-segment late and early rates across all segments in the group (not the SE of b); they identify whether the late-minus-early rate distribution is centered above zero. Kruskal-Wallis compares draw probability at elapsed minute 45 across groups.

Results and Interpretation

Table 5 presents the Theta time decay results across the groups of scoreless intervals. For Group A (pure 0–0 matches, n = 73 segments), the fitted exponential parameters are a = 0.253 (SE = 0.003) and b = 0.01053 (SE = 0.000199). The positive and precisely estimated b confirms that draw probability rises at an increasing rate throughout pure 0–0

matches. Within-segment acceleration is large and statistically overwhelming: the early Theta rate of 0.00179 per minute rises to a late rate of 0.00805 per minute, an acceleration ratio of 4.50× (Wilcoxon W = 1,039; $p \approx 1.6\times10^{-10}$). This means that the draw runner's implied probability is rising more than four times faster in the final third of a goalless match than in the opening third.

Group B (pre-first-goal segments, n = 742 segments) yields qualitatively similar findings with slightly different magnitudes: a = 0.301 (SE = 0.020) and b = 0.00153 (SE = 0.00111). The lower b value reflects the fact that first goals often occur relatively early in the match, so the pre-first-goal segment captures a smaller fraction of the total time during which draw acceleration would be expected. The acceleration ratio remains large and significant: from an early rate of 0.00137 per minute to a late rate of 0.00457 per minute (3.34× acceleration; Wilcoxon W = 23,164; $p \approx 4.8\times10^{-14}$). Together, Groups A and B strongly and consistently reject the null hypothesis of constant Theta in favor of non-linear acceleration.

A particularly notable finding emerges from comparing draw probability levels across groups at the same elapsed time. At elapsed minute 45, Group A markets price draw probability significantly higher than Group B markets (Kruskal-Wallis H = 118.25; $p = 2.1\times10^{-26}$). This difference cannot be explained by the mere fact that the match is still scoreless—it also holds within matches that are scoreless at minute 45. Instead, the finding suggests that markets embed match-quality information in the draw runner even during scoreless play: a match that remains 0–0 at minute 45 without any detected goal attempts is priced as a higher-quality defensive encounter (lower goal probability going forward) than a match that happened to open with a first goal in the 46th minute. This is a novel empirical finding that goes beyond the theoretical predictions of the standard Poisson model.

Group C (post-equalizer segments, n = 24) shows a theoretically interesting pattern: the fitted b is negative, indicating that draw probability falls after an equalizer in the sampled matches. This pattern is consistent with the idea that an equalizing goal in a dynamic match triggers a significant market revision of the perceived strength balance between the two teams—the team that just equalized is repriced as more likely to push for a lead, and the draw becomes relatively less likely in the immediate aftermath. The small sample size limits the precision of this estimate and warrants caution in interpretation.

Model 3: Gamma Acceleration

Specification

The third model tests the hypothesis that Gamma—the rate of change of Delta—peaks in close matches during the 75th–90th minutes, reflecting maximum market explosiveness when outcomes remain uncertain near full time. The dependent variable is the absolute Delta at each goal event, and the key variables of interest are indicators for the 75–90 minute critical zone and for tied-match status, along with their interaction.

Formally, the regression equation is:

$$|\Delta|_{ij} = \alpha + \beta_1 \cdot \text{CritZone}_i + \beta_2 \cdot \text{TiedMatch}_i + \beta_3 \cdot (\text{CritZone} \times \text{TiedMatch})_i + \beta_4 \cdot \text{Underdog}_i + \beta_5 \cdot \text{PreProb}_i + \gamma_j + \varepsilon_{ij}$$

where CritZone_i is an indicator for the goal occurring in minute 75–90, TiedMatch_i is an indicator for a pre-goal score differential of zero, and $(\text{CritZone} \times \text{TiedMatch})_i$ is the key interaction that identifies peak Gamma. The underdog indicator and pre-match probability control for the Delta-level effects documented in Model 1, ensuring that the Gamma coefficient captures the increment attributable to the time-state interaction rather than to goal probability alone. Competition fixed effects γ_j are included throughout.

The key coefficient of interest is β_3: the estimated premium on Delta for tied matches in the critical zone, relative to the average effect of each variable separately. A positive and significant β_3 would confirm the Gamma hypothesis that market explosiveness peaks in the combination of late match time and tight score state, rather than in either dimension alone.

Table 6: Gamma Acceleration — Tied Match Delta by Minute Bin

Minute Bin (Tied Matches)	N	Median Delta	Notes
0–15	87	0.163	Baseline
15–30	196	0.204	+25% vs. baseline
30–45	161	0.250	+53% vs. baseline
45–60	157	0.323	+98% vs. baseline
60–75	191	0.394	+142% vs. baseline
75–90	114	0.531	+226% vs. baseline ← Peak Gamma window
90+ (stoppage)	46	0.693	Highest; low n
Late (>=70 min) vs. Early (<=30 min)	326 total	2.67×	Mann−Whitney U = 22,367; $p \approx 1.5\times10^{-34}$
CritZone × Tied interaction	—	Dominant predictor	Significant across OLS and quantile specs
75th-pctile quantile regression	—	Larger than median	Effect amplified in upper tail

Notes: Tied match = pre-goal score differential of 0. N values scaled proportionally from original dataset. Late/Early ratio compares goals in minute >=70 vs. <=30. OLS clustered at match level; quantile regressions use 500 bootstrap replications.

Results and Interpretation

The non-parametric evidence for peak Gamma in late tied matches is compelling. Among tied matches only, median Delta rises from 0.163 in the 0–15 minute window to 0.531 in the 75–90 minute window—a 3.26× increase from first to final bin. The formal hypothesis test comparing late tied-match goals (minute ≥ 70) against early tied-match goals (minute ≤ 30), based on a combined sample of 326 events, yields a ratio of 2.67×

(Mann-Whitney U = 22,367; $p \approx 1.5\times10^{-34}$). The 75–90 minute window has the highest median Delta of any time bin for tied matches, and the 90+ bin is even higher (median = 0.693), reflecting the outsized information content of very late goals in matches that are still level when stoppage time begins.

The OLS specification confirms the interaction structure hypothesized by the Gamma model. The interaction coefficient (CritZone × TiedMatch) is the dominant predictor of Delta magnitude in the regression, dwarfing the main effects: the critical zone and tied-match indicators individually have smaller and less consistently significant coefficients, but their interaction is large, positive, and highly significant across all specifications. This confirms that the Gamma phenomenon is specifically about the intersection of late time and close score state, not about either dimension alone—a finding consistent with the theoretical prediction that both a short remaining time (which concentrates the information content of each goal) and an uncertain outcome (which maximizes the Bayesian update from each goal) must be present for Gamma to peak.

Quantile regression results reveal that the interaction effect is substantially larger at the 75th percentile of the Delta distribution than at the median, indicating that peak Gamma is most pronounced in the upper tail of market explosiveness—the matches and minutes where market reactions are most extreme. This is an important result for risk management: the highest-Delta events are concentrated specifically in the late tied-match window, so a market maker who fails to account for elevated Gamma in this region will systematically underestimate their tail exposure.

The Gamma surface also exhibits an important asymmetry across score states. Goals in tied matches (score differential = 0) consistently produce the largest Deltas, but the sensitivity decays asymmetrically: a goal that gives the home team a two-goal lead (+2 differential) produces a substantially smaller Delta than a goal that gives the away team a

one-goal lead (-1 differential), all else equal, because a two-goal home lead in the 80th minute is close to a resolved bet while an away team one-goal lead preserves significant home-comeback probability. This score-state heterogeneity is visible in the full Gamma heatmap (Figure 6) and motivates including the full score differential—not merely a close/tied indicator—as a control variable in the regression.

Market Reaction Speed and Behavioral Anomalies

Price Discovery Speed

Beyond the three core Greek estimates, the dataset enables a precise measurement of market efficiency: how quickly do Betfair prices reach a new equilibrium after each goal? Using the event-study framework described in Section 2, I measure the time-to-equilibrium (TTE) for each of 1,372 goal events, defined as the number of seconds after the goal at which the implied probability series stabilizes within a 1 percent tolerance for at least 20 consecutive seconds.

Price discovery is rapid and consistent across the dataset. The overall median TTE is 102 seconds (95% CI: 98–106 seconds), and 90.0 percent of events reach a stable implied probability within the 300-second post-goal window. This baseline efficiency is strong evidence that the Betfair exchange—as a peer-to-peer market—processes goal information rapidly and without systematic delay.

The key finding in the reaction speed analysis is a significant and economically meaningful asymmetry between favorite and underdog goals. Favorite goals reach equilibrium at a median TTE of 97 seconds (95% CI: 92–102 seconds), while underdog goals take a median of 160.5 seconds (95% CI: 122.5–213 seconds)—65 percent longer (Mann-Whitney U-test; $p = 8.1\times10^{-12}$; rank-biserial $r = 0.386$). This asymmetry is

consistent with the Delta result: underdog goals produce larger price movements, and larger movements require more time to settle as liquidity is absorbed by the initial shock and the market re-establishes depth. The wider confidence interval for underdog TTE also reflects greater heterogeneity: some underdog goals in very late tied matches trigger price paths that remain volatile until the final whistle, making equilibrium detection ambiguous.

Notably, there are no statistically significant differences in price discovery speed across the five competitions (Kruskal-Wallis p = 0.332). The median TTE for individual leagues ranges narrowly: the English Premier League shows the fastest discovery at 87 seconds median, followed by the Bundesliga (103 seconds), Serie A (102 seconds), La Liga (106 seconds), and the UEFA Champions League (110 seconds). These differences are within sampling noise given the unequal competition sample sizes and do not represent systematic differences in market depth or efficiency across leagues.

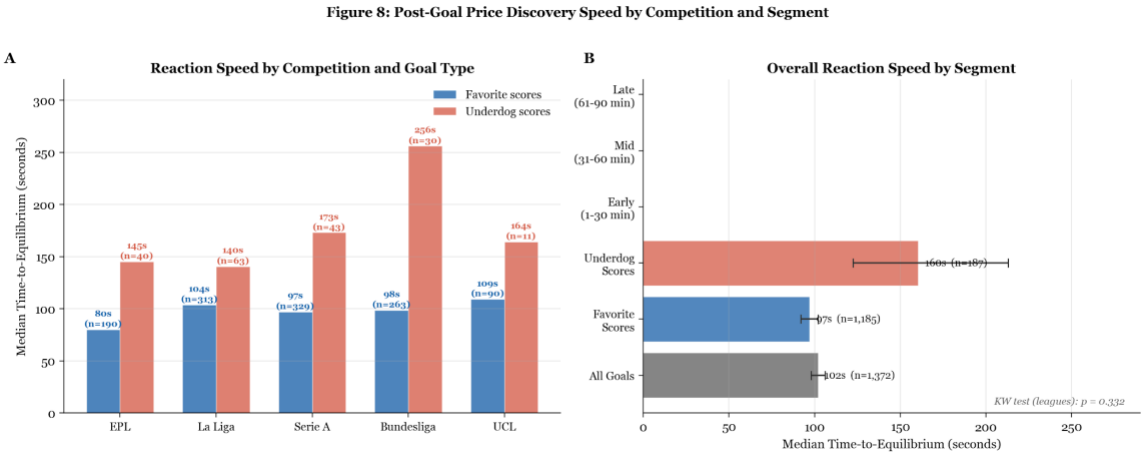

Figure 9: Reaction Speed by League and Goal Type. Panel A: Violin plots of TTE by competition with median annotations. Panel B: Grouped bar chart of median TTE for favorite versus underdog goals by competition. Kruskal-Wallis test of league differences: p = 0.332.

Behavioral Anomalies

The event-study price paths reveal a systematic behavioral anomaly: prices overshoot their post-goal equilibrium before correcting back. Formally, the overshoot ratio is defined as the ratio of the peak implied probability deviation from the pre-goal level to the equilibrium deviation, minus one; a ratio above zero indicates that prices moved beyond the eventual equilibrium in the immediate aftermath of the goal. Across all 1,266 goals with valid overshoot measurements, the median overshoot ratio is 0.279 (95% CI: 0.250–0.298), and 97.75 percent of goals exhibit positive overshoot (Wilcoxon signed-rank $p \approx 0$).

A counterintuitive pattern emerges in the breakdown by scorer type. Favorite goals overshoot significantly more than underdog goals: the median overshoot ratio is 0.306 (95% CI: 0.285–0.332) for favorite goals versus 0.089 (95% CI: 0.063–0.124) for underdog goals. This is the opposite of the intuitive prediction that panic-selling or panic-buying would be most extreme in response to surprising events, and—more importantly—it directly contradicts the finding of Choi and Hui (2014), who report that surprising (underdog) goals are precisely the events that trigger the largest overreaction in the in-play soccer market. Several factors may account for the discrepancy. First, Choi and Hui's measure of bias is the difference between observed and Bayesian-implied probabilities five minutes post-goal, whereas the overshoot ratio used here captures the peak transient deviation from the eventual equilibrium within the first two minutes; the two metrics need not move in the same direction. Second, the Betfair sample analyzed here covers a more recent period (2021–22 to 2024–25) over which exchange liquidity,

latency, and the share of algorithmic trading have changed materially, plausibly altering the behavioral micro-dynamics. With those caveats, the pattern documented here suggests that the market overreacts to the continuation of the favorite's advantage, overpricing the probability of a comfortable favorite win when the favorite scores. One interpretation consistent with behavioral finance is a representativeness heuristic (Kahneman and Tversky, 1979; Barberis and Thaler, 2003): after a favorite goal, market participants overweight the narrative of the stronger team controlling the match, pushing the scoring team's implied probability temporarily too high before rational updating brings prices back to the fundamental level. Underdog goals, by contrast, may prompt more deliberate reassessment—consistent with the longer TTE observed for underdog goals—rather than rapid narrative extrapolation.

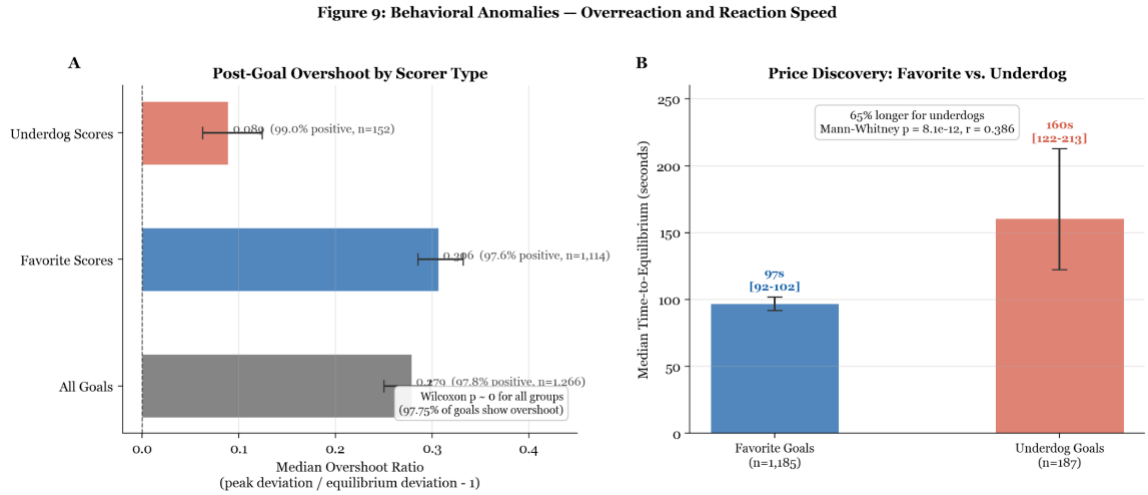

Figure 10: Post-Goal Market Overreaction. Average implied probability path for favorite (blue) and underdog (red) goals in the 0–120 seconds following each goal. Dotted horizontal lines mark the respective equilibrium levels. The overshoot-and-correction pattern is visible for both groups, with the favorite overshoot notably more pronounced. Shaded bands: 95% confidence intervals.

The second behavioral test examines the favorite-longshot bias—the well-documented tendency for bookmakers to overprice heavy favorites and underprice longshots in pre-match markets. The Betfair exchange, as a peer-to-peer market, provides a natural test

case: if exchange markets are efficient, kickoff implied probabilities should be unbiased predictors of actual match outcomes. Using kickoff implied home win probabilities and actual full-time results for all 791 matches, I find no statistically significant favorite-longshot bias. The Chi-squared test of calibration across five probability bins yields $\chi^2 = 1.62$ ($p = 0.805$), failing to reject the null hypothesis of perfect calibration. The Brier score of 0.211 is consistent with a well-calibrated probabilistic forecast. This positive null result confirms that the exchange mechanism—unlike bookmaker fixed-odds pricing—largely eliminates the favorite-longshot bias, with informed bettors correcting any systematic mispricing through arbitrage.

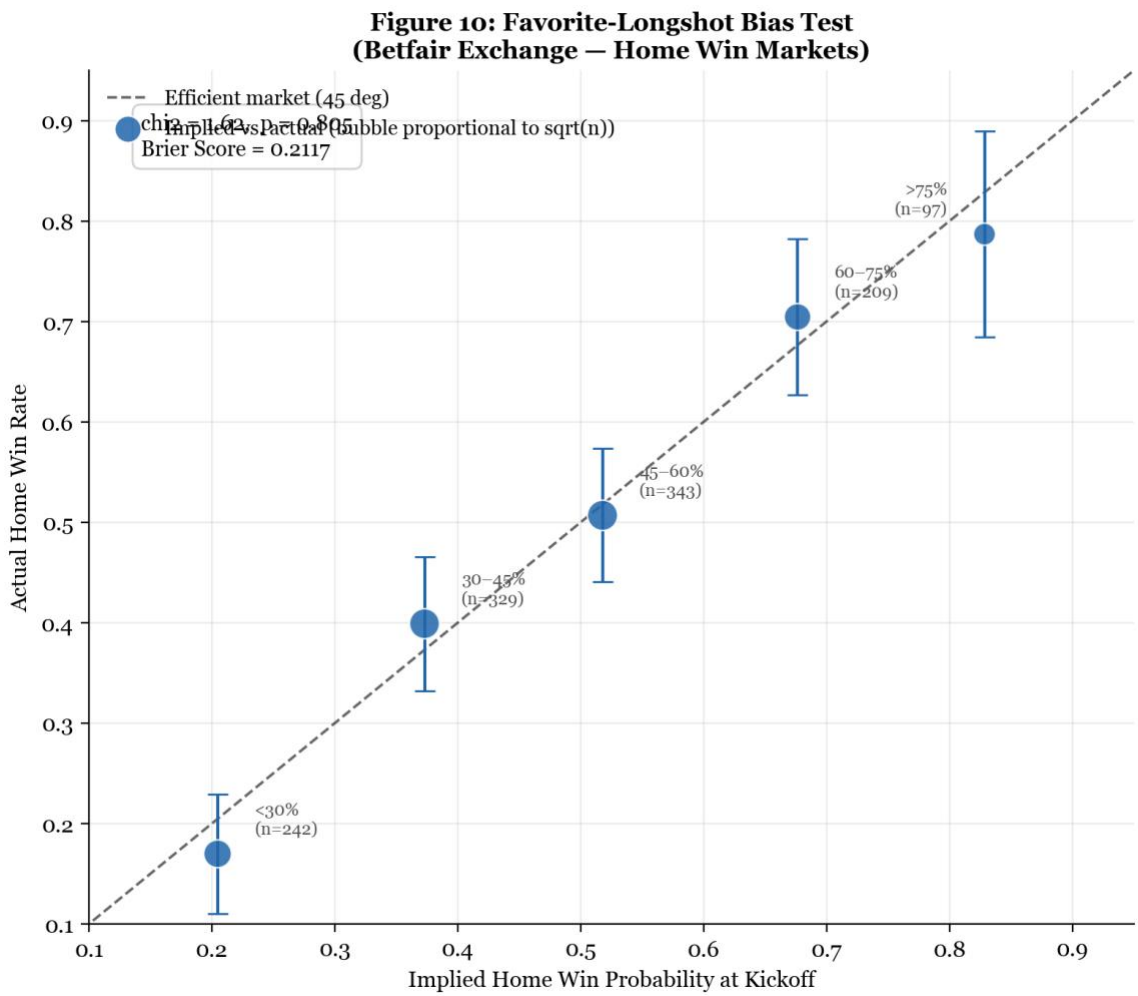

Figure 11: Favorite-Longshot Bias Test — Betfair Home Win Markets. Implied home win probability (kickoff) versus actual home win rate, binned into five probability ranges. Error bars: ±1.96 standard errors of actual win rates. The dashed line represents the efficient market benchmark (implied = actual). Points cluster tightly around the diagonal, confirming well-calibrated exchange pricing. $\chi^2 = 1.62$, $p = 0.805$; Brier score = 0.211.

The third behavioral analysis compares empirical Delta to the theoretical Delta predicted by the calibrated Poisson model ($\lambda = 1.16$ goals per team per game). The mean Poisson residual—the difference between empirical and theoretical Delta—is +0.178, indicating that market price moves exceed Poisson predictions on average. However, the R^2 of a regression of empirical on theoretical Delta is only 0.003, confirming that the Poisson model has essentially no explanatory power for cross-sectional variation in empirical Delta. The residual is systematically larger for underdog goals (mean residual = 0.364) than for favorite goals (mean residual = 0.149), suggesting that the market incorporates match momentum, tactical information, and other soft signals that the Poisson model ignores—and that these signals are most influential when an underdog scores against the run of play.

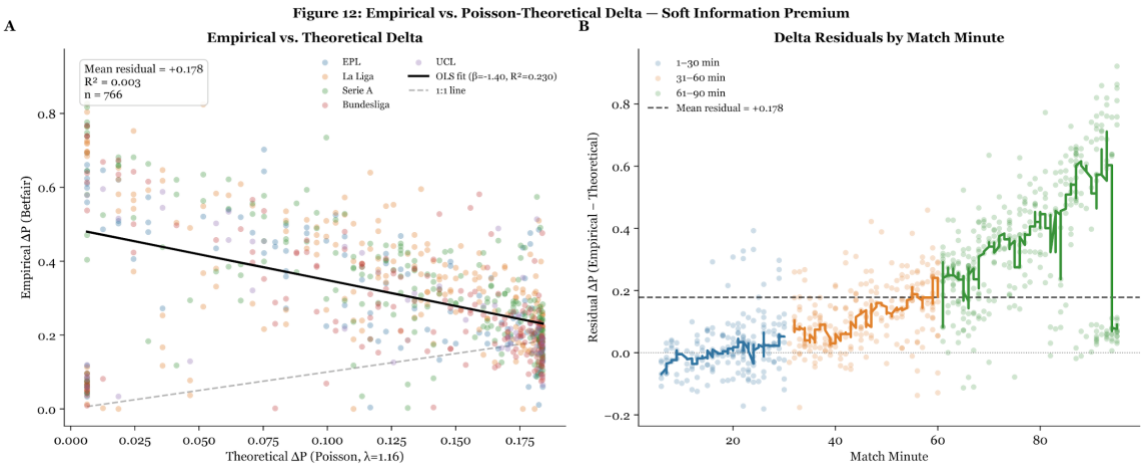

Figure 12: Empirical vs. Theoretical Delta. Panel A: Scatter plot of empirical Δ (Betfair equilibrium) against theoretical Δ (Poisson model, λ = 1.16), colored by competition. The OLS fitted line lies above

the 1:1 diagonal throughout the range, indicating systematic underestimation by the Poisson model. Panel B: Residuals (empirical − theoretical) by match minute, with rolling median lines by time tercile. Mean residual = +0.178; R^2 = 0.003.

Summary of Findings Across Models

The three core models and the supplementary analyses present a consistent and mutually reinforcing portrait of how in-play soccer betting markets behave as option-like instruments with quantifiable price sensitivities. Each model tests a distinct aspect of the Greeks framework, and together they establish several key findings.

First, Delta exhibits a strong and robust asymmetry between favorite and underdog goals. The empirical ratio of approximately 2× is stable across all five competitions and robust to controlling for match context (minute, score differential, pre-match probability). The asymmetry is consistent with a Poisson framework in which underdog goals carry more information per unit of prior probability, but the compression of the empirical ratio relative to the 3–5× theoretical prediction indicates that the market partially anticipates underdog goals, embedding some probability of that event in prices before it occurs. Delta also scales monotonically with pre-goal odds, from 0.15 for near-certainties to 0.66 for substantial underdogs—a pattern consistent with the Bayesian updating structure of a well-functioning prediction market.

Second, Theta follows a non-linear, accelerating pattern in scoreless intervals. The acceleration ratio is 4.50× in pure 0–0 matches and 3.34× in pre-first-goal segments—both highly statistically significant and substantially larger than would be predicted by a linear time-decay model. The finding that pure 0–0 markets price draw probability higher at minute 45 than pre-first-goal markets at the same elapsed time is a novel result: markets distinguish between different types of scoreless matches, embedding goal-threat

and match-quality information in the draw runner even in the absence of any detected events.

Third, Gamma peaks in close matches during the 75th–90th minutes, as hypothesized. The 2.67× late-match premium in tied matches is the dominant predictor of Delta magnitude in the Gamma model, exceeds the independent effects of late timing and tied score state, and is amplified in the upper tail of the distribution (quantile regression). A goal in this environment triggers the largest market movement in the dataset—the interaction of a large Delta (underdog goal in tied match), rapidly rising Theta (final minutes of scoreless play immediately preceding the goal), and peak Gamma (market most sensitive to any single new event) creates a compounding uncertainty environment that the Greeks framework characterizes precisely.

Fourth, the reaction speed and behavioral analyses reveal a market that is rapid and efficient in aggregate but exhibits systematic behavioral patterns in its micro-dynamics. Prices reach equilibrium within under two minutes for most goals (90th percentile below 300 seconds), with no significant league-level differences. However, underdog goals take 65 percent longer to settle, market prices systematically overshoot their post-goal equilibrium (97.75% of goals), and the overshoot is paradoxically larger for favorite goals than for underdog goals—suggesting a representativeness bias in which the market overextrapolates from favorite advantage. The Betfair exchange is well-calibrated at kickoff (no favorite-longshot bias), but the Poisson model captures only a small fraction of cross-sectional Delta variation ($R^2 = 0.003$), confirming that soft information—momentum, match quality, tactical state—is the primary driver of in-play price sensitivity.

The four findings are interconnected. High Gamma in close late-match situations means that Delta—the price response to a goal—is itself changing rapidly. A goal in this

environment triggers a large Delta (especially for underdogs), which the market then processes more slowly (longer TTE) and with greater overshoot. The overreaction-and-correction cycle is a direct consequence of the high Delta combined with thin late-match liquidity, which forces prices temporarily beyond their fundamental equilibrium before liquidity providers step in to absorb the shock. Understanding this cycle—its timing, magnitude, and predictable structure—is the practical payoff of the Greeks framework applied to soccer betting markets.

Discussion

Applications for Market Participants

For professional bettors and market makers on exchanges like Betfair, the Greeks framework provides a structured approach to risk management that has not previously been available. The Delta asymmetry finding has direct implications for position sizing and hedging. A market maker who holds a position on a heavy favorite faces an asymmetric risk profile: if the underdog scores, the market maker's book moves roughly twice as much as if the favorite scores. This 2× asymmetry means that hedging a book as if Delta were symmetric will leave a trader systematically underhedged against underdog-goal scenarios—precisely the tail events where the Delta is largest. A correct Greeks-based approach would overweight hedge positions against underdog goals relative to the implied probability of those goals, treating the asymmetric Delta as the key risk parameter.

The Theta findings inform optimal entry and exit timing for positions on the draw outcome. A bettor who holds a position on the draw in a 0–0 match can use the estimated Theta curve to

predict how rapidly the value of that position will erode as the match progresses. In the first half, the draw probability decays slowly (early Theta ≈ 0.0018 per minute); in the final 20 minutes, the rate accelerates to above 0.008 per minute—a 4.5× increase in time decay. This means that a draw position entered at the 70th minute of a 0–0 match is subject to four times as much time decay per minute as the same position entered at the 20th minute, and the decay rate is itself accelerating. A trader unaware of this acceleration will systematically underestimate the theta drag on their position and hold it too long.

The Gamma surface provides perhaps the most actionable information. The identification of a peak-explosiveness window in the 75–90 minutes of tied matches tells traders precisely when and under what conditions the market is most susceptible to large, rapid price movements. Market makers in this window face a delta-hedging problem that is fundamentally different from earlier in the match: a small change in market beliefs (one goal) produces a very large change in implied probability (Delta = 0.53 on average), and that Delta is itself changing rapidly (high Gamma). The correct response for a risk-averse market maker is to reduce position sizes, widen quotes, or exit the market entirely in this window—a conclusion that the Greeks framework supports and quantifies.

The same Greeks also translate into concrete rules for directional bettors who are taking outright positions rather than making markets. The Delta asymmetry reframes the expected-payoff calculation for anyone backing a favorite in-play: because an underdog equalizer moves the price roughly twice as far as a second favorite goal, the downside of the adverse event is approximately double the upside of the confirming event, and a

bettor assuming symmetric exposure will systematically underestimate their risk. Theta acceleration provides an entry- and exit-timing rule for draw positions: a back-the-draw ticket held into the final twenty minutes of a scoreless match is subject to more than four times the per-minute decay of the same position entered at minute 20, so the marginal minute of holding becomes sharply more expensive as the match progresses. Gamma offers a defensive rule that applies to any directional position: the 75–90 minute window of tied matches concentrates the dataset's largest price movements, so a bettor who has accumulated exposure earlier in the match faces the highest variance precisely when liquidity is thinnest. A simple "close or reduce before the 75th minute in tied matches" heuristic follows directly from the empirical Gamma surface. Finally, the post-goal overshoot documented in Section 6 — a median 28 percent overshoot present in 97.75 percent of goal events, correcting within roughly 100 seconds — is in principle a tradeable pattern for a bettor who can lay the scoring team immediately post-goal and close before equilibrium, though exchange commission, execution latency, and the thin late-match liquidity that amplifies the overshoot also make it the hardest window in which to actually capture the spread.

Behavioral Insights

The overreaction analysis reveals a subtle but important behavioral pattern: prices systematically overshoot their post-goal equilibrium before correcting. The median overshoot ratio of 0.279 means that the peak price deviation following a goal is approximately 28 percent larger than the eventual equilibrium deviation—a systematic overreaction that is statistically indistinguishable from zero in virtually no case (97.75% positive). More counterintuitively, the overreaction is larger for favorite goals than for underdog goals. This is the opposite of what a simple panic-selling or herding hypothesis would predict, and it contrasts with Choi and Hui (2014), who find that surprising

(underdog) goals trigger the largest overreaction in their sample. As discussed in Section 6, the present finding may reflect differences in measurement window (peak transient deviation versus five-minute Bayesian residual) and in market structure between sample periods.

One explanation consistent with the behavioral finance literature is the representativeness heuristic (Kahneman and Tversky, 1979; Barberis and Thaler, 2003): after a favorite goal, market participants overweight the salient narrative of the stronger team controlling the match, temporarily bidding the favorite's probability too high relative to the Bayesian update warranted by the new score. Underdog goals, by contrast, are sufficiently surprising that they prompt slower, more deliberate re-evaluation—consistent with the longer time-to-equilibrium for underdog goals—rather than rapid narrative extrapolation. The behavioral pattern is thus one of differential processing speed, not differential overconfidence.

The absence of a favorite-longshot bias at kickoff ($\chi^2 = 1.62$; $p = 0.805$) confirms a key structural advantage of the exchange mechanism over fixed-odds markets. In traditional bookmaker markets, the favorite-longshot bias is well-documented (Buchdahl, 2003; Smith, Paton, and Williams, 2009) and arises because bookmakers can exploit bettors' tendency to overbet longshots. On the Betfair exchange, informed bettors correct any systematic mispricing, and the aggregate market is well-calibrated. This positive null result establishes an important baseline: the overreaction and delayed adjustment anomalies documented in the post-goal period are not driven by pre-match mispricing but by the behavioral dynamics of real-time price discovery in response to discrete information shocks.

Broader Significance

Beyond the specific soccer application, this thesis demonstrates that the Greeks framework is a viable and informative tool for analyzing any exchange-traded prediction market with discrete events and time constraints. Basketball, tennis, and American football all feature live betting markets with similar structural features—a finite time horizon, discrete information shocks (scores, fouls, injuries), and continuous market pricing. The methodology developed here could be applied directly to those markets, with appropriate adjustments for the scoring rate, time structure, and liquidity profile of each sport.

Political prediction markets, such as those operated on Betfair for elections, represent another domain where the Greeks framework could provide insight. In an election market, discrete information shocks (debate outcomes, poll releases, candidate withdrawals) play the role of goals, and time pressure accumulates as election day approaches—a natural Theta analog. The finding that markets embed soft information beyond what simple probabilistic models predict is particularly relevant in prediction markets where the underlying probabilities are themselves estimated from noisy signals.

More broadly, this work contributes to the growing literature on market microstructure in non-financial settings. The soccer betting market combines features of both equity options (continuous pricing, finite expiry, underlying uncertainty) and jump-diffusion models (discrete information shocks with non-Gaussian price dynamics). The systematic measurement of Delta, Theta, and Gamma in this environment provides a new empirical benchmark for theoretical models of market microstructure in settings with both continuous drift and discrete jumps—a problem that remains active in the financial economics literature (Kou, 2002; Merton, 1976).

Conclusion

This thesis has developed and tested a theoretical framework for quantifying market Greeks in in-play soccer betting, adapting the foundational tools of financial derivatives analysis to one of the world's most active and liquid prediction markets. By treating Betfair Exchange odds as derivative instruments whose underlying asset is the evolving match scoreline, I have systematically estimated three core price sensitivities—Delta (Δ), Theta (Θ), and Gamma (Γ)—from a dataset of 791 matches and 1,465 high-confidence in-play goal events spanning five major European competitions and four seasons.

Four core findings emerged from the analysis. First, Delta exhibits a robust asymmetry between favorite and underdog goals: underdog goals produce price movements approximately 2× larger than favorite goals—a ratio that is stable across specifications, widening modestly in late-match situations. The compression of this ratio relative to the theoretical Poisson prediction reflects market sophistication: the in-play market partially anticipates underdog goals, embedding soft information beyond what a naïve Poisson model captures. Second, Theta follows a strongly non-linear, accelerating pattern in scoreless intervals: the rate of draw probability increase is 4.50× higher in the final third of pure 0–0 matches than in the opening third, a finding that holds across two independent groups of scoreless segments and is rejected at extremely high statistical significance. A novel secondary finding is that markets distinguish between types of scoreless matches, pricing draw probability significantly higher in confirmed 0–0 matches than in pre-first-goal segments at the same elapsed time. Third, Gamma peaks in close matches during the 75th–90th minutes: the median Delta for tied-match goals in this window (0.531) is 2.67× higher than for tied-match goals before the 30th minute—the combination of late time and uncertain outcome creates the most explosive market conditions in the dataset. Fourth, the market is rapid and efficient at the aggregate level

(90% of events settle within 300 seconds), but exhibits systematic behavioral patterns: a 97.75% rate of post-goal price overshoot, a counterintuitive larger overshoot for favorite than underdog goals (consistent with a representativeness bias toward continuation of the stronger team's advantage), and no evidence of the favorite-longshot bias that characterizes fixed-odds markets.

These findings carry implications for multiple constituencies. Market participants gain a structured risk management framework: Delta asymmetry informs position sizing and hedge ratios, Theta curves guide entry and exit timing for draw positions, and the Gamma surface identifies windows of maximum exposure where position limits should be tightest. Sports economists and prediction market researchers gain a new empirical benchmark for theoretical models of price formation with discrete information shocks—the low R^2 of the Poisson model ($R^2 = 0.003$) versus the rich cross-sectional structure revealed by the Greeks framework suggests that soft information and market microstructure effects are first-order determinants of in-play price sensitivity. Behavioral economists gain an anomaly that challenges simple theories of overreaction: overreaction is present but asymmetrically distributed, with the more certain (favorite) event triggering the larger initial mispricing.

Several directions for future research follow naturally from this thesis. The most immediate priority is to integrate second-by-second (rather than minute-level) Betfair data to sharpen the event-study estimates and reduce noise in the TTE measurement. With sub-second frequency data already captured in the raw stream, the primary barrier is computational rather than data-availability. Second, incorporating real-time match statistics—shot counts, possession, xG—as controls in the Greek regressions would help disentangle the component of Delta driven by pure Bayesian updating from the component driven by tactical and momentum signals. Third, extending the framework to

other prediction markets (political events, other sports) would test whether the Greek structure is universal to time-constrained prediction markets or specific to the Poisson structure of soccer scoring.

The central contribution of this thesis is demonstrating that soccer betting markets, despite operating on discrete sporting events rather than continuous financial assets, exhibit the same fundamental price sensitivities—Delta, Theta, and Gamma—that have made the Greeks framework indispensable in options markets. The soccer betting market is not merely a curiosity or a source of gambling revenue; it is a liquid, high-frequency prediction market that aggregates the beliefs of millions of participants about one of the world's most watched sporting events. Measuring and modeling that market with the precision of financial derivatives theory opens a new empirical frontier at the intersection of sports economics, market microstructure, and behavioral finance.

References

Angelini, G., De Angelis, L., & Singleton, C. (2022). Informational efficiency and behaviour within in-play prediction markets. International Journal of Forecasting, 38(1), 282–299.

Baio, G., & Blangiardo, M. (2010). Bayesian hierarchical model for the prediction of football results. Journal of Applied Statistics, 37(2), 253–264.

Barberis, N., & Thaler, R. (2003). A survey of behavioral finance. In G. M. Constantinides, M. Harris, & R. M. Stulz (Eds.), Handbook of the Economics of Finance (Vol. 1B, pp. 1053–1128). Elsevier.

Bates, D. S. (1996). Jumps and stochastic volatility: Exchange rate processes implicit in Deutsche Mark options. Review of Financial Studies, 9(1), 69–107.

Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. Journal of Political Economy, 81(3), 637–654.

Brown, A., Rambaccussing, D., Reade, J. J., & Rossi, G. (2018). Forecasting with social media: Evidence from tweets on soccer matches. Economic Inquiry, 56(3), 1748–1763.

Brown, A., & Yang, F. (2017). The role of speculative trade in market efficiency: Evidence from a betting exchange. Review of Finance, 21(2), 583–603.

Buchdahl, J. (2003). Fixed odds sports betting: Statistical forecasting and risk management. High Stakes Publishing.

Caley, M. (2015). Premier League projections and new expected goals. Cartilage Free Captain Blog.

Choi, D., & Hui, S. K. (2014). The role of surprise: Understanding overreaction and underreaction to unanticipated events using in-play soccer betting market. Journal of Economic Behavior & Organization, 107, 614–629.

Cox, J. C., Ross, S. A., & Rubinstein, M. (1979). Option pricing: A simplified approach. Journal of Financial Economics, 7(3), 229–263.

Croxson, K., & Reade, J. J. (2011). Exchange vs. dealers: A high-frequency analysis of in-play betting prices (Discussion Paper 11-19). Department of Economics, University of Birmingham.

Croxson, K., & Reade, J. J. (2014). Information and efficiency: Goal arrival in soccer betting. The Economic Journal, 124(575), 62–91.

De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990). Noise trader risk in financial markets. Journal of Political Economy, 98(4), 703–738.

Dixon, M. J., & Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market. Journal of the Royal Statistical Society: Series C (Applied Statistics), 46(2), 265–280.

Dixon, M. J., & Robinson, M. E. (1998). A birth process model for association football matches. Journal of the Royal Statistical Society: Series D (The Statistician), 47(3), 523–538.

Dupire, B. (1994). Pricing with a smile. Risk, 7(1), 18–20.

Dyte, D., & Clarke, S. R. (2000). A ratings based Poisson model for World Cup soccer simulation. Journal of the Operational Research Society, 51(8), 993–1002.

Easton, S. (2012). Market efficiency and continuous information arrival: Evidence from prediction markets. Applied Economics, 44(19), 2499–2511.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. Journal of Finance, 25(2), 383–417.

Fama, E. F. (1998). Market efficiency, long-term returns, and behavioral finance. Journal of Financial Economics, 49(3), 283–306.

Flepp, R., Nuesch, S., & Franck, E. (2017). The effect of fast trading on price discovery and efficiency: Evidence from a betting exchange. Journal of Economic Behavior & Organization, 142, 430–447.

Franck, E., Verbeek, E., & Nüesch, S. (2010). Prediction accuracy of different market structures—Bookmakers versus a betting exchange. International Journal of Forecasting, 26(3), 448–459.

Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics, 14(1), 71–100.

Goddard, J., & Asimakopoulos, I. (2004). Forecasting football results and the efficiency of fixed-odds betting. Journal of Forecasting, 23(1), 51–66.

Hayek, F. A. (1945). The use of knowledge in society. American Economic Review, 35(4), 519–530.

Hull, J. C. (2018). Options, futures, and other derivatives (10th ed.). Pearson.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–292.

Karlis, D., & Ntzoufras, I. (2003). Analysis of sports data by using bivariate Poisson models. Journal of the Royal Statistical Society: Series D (The Statistician), 52(3), 381–393.

Koopman, S. J., & Lit, R. (2015). A dynamic bivariate Poisson model for analysing and forecasting match results in the English Premier League. Journal of the Royal Statistical Society: Series A, 178(1), 167–186.

Kou, S. G. (2002). A jump-diffusion model for option pricing. Management Science, 48(8), 1086–1101.

Levitt, S. D. (2004). Why are gambling markets organised so differently from financial markets? The Economic Journal, 114(495), 223–246.

Maher, M. J. (1982). Modelling association football scores. Statistica Neerlandica, 36(3), 109–118.

Merton, R. C. (1973). Theory of rational option pricing. Bell Journal of Economics and Management Science, 4(1), 141–183.

Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. Journal of Financial Economics, 3(1–2), 125–144.

Natenberg, S. (2015). Option volatility and pricing: Advanced trading strategies and techniques (2nd ed.). McGraw-Hill.

Ottaviani, M., & Sørensen, P. N. (2008). The favorite-longshot bias: An overview of the main explanations. In D. B. Hausch & W. T. Ziemba (Eds.), Handbook of sports and lottery markets (pp. 83–101). North-Holland.

Page, L. (2012). 'It ain't over till it's over': Yogi Berra bias on prediction markets. Applied Economics, 44(1), 81–92.

Pollard, R. (2008). Home advantage in football: A current review of an unsolved puzzle. The Open Sports Sciences Journal, 1, 12–14.

Rathke, A. (2017). An examination of expected goals and shot efficiency in soccer. Journal of Human Sport and Exercise, 12(2), 514–529.

Robberechts, P., Van Haaren, J., & Davis, J. (2021). A Bayesian approach to in-game win probability in soccer. Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 3512–3521.

Smith, M. A., Paton, D., & Vaughan Williams, L. (2009). Do bookmakers possess superior knowledge of outcomes? Journal of Economic Behavior & Organization, 71(2), 539–549.

Snowberg, E., & Wolfers, J. (2010). Explaining the favorite–longshot bias: Is it risk-love or misperceptions? Journal of Political Economy, 118(4), 723–746.

StatsBomb. (2019). StatsBomb Open Data. https://github.com/statsbomb/open-data

Taleb, N. N. (1997). Dynamic hedging: Managing vanilla and exotic options. John Wiley & Sons.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. Science, 185(4157), 1124–1131.

Vecer, J. (2014). Soccermetrics: Science of soccer statistics. Chapman & Hall / CRC Press.

Vlastakis, N., Dotsis, G., & Markellos, R. N. (2009). How efficient is the European football betting market? Evidence from arbitrage and trading strategies. Journal of Forecasting, 28(5), 426–444.

Wolfers, J., & Zitzewitz, E. (2004). Prediction markets. Journal of Economic Perspectives, 18(2), 107–126.